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The Classical Theory of Zeta and L-Functions

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Abstract. This expository paper is devoted to basic facts about Zeta and L-functions, following to some extent the historical development of the subject.

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1. Early history

Euler's father wanted his son to follow him into the church and sent him to the University of Basel to prepare for the ministry. He entered the University in 1720, at the age of 14. He began his study of theology in 1723 but, although he was to be a devout Christian all his life, his enthusiasm was for mathematics. He obtained his father's consent to change to mathematics after Johann Bernoulli persuaded him of his son's exceptional mathematical talent.

After Basel he went to St. Petersbourg in 1727, then back to Basel in 1733 where he married Katharina Gsell. They had 13 children, but only five survived. Euler claimed that he made some of his greatest mathematical discoveries while holding a baby in his arms with other children playing round his feet. After a period in Berlin, that terminated after disputes with Maupertuis in the Berlin Academy on the principle of minimum action, he went back to St. Petersbourg in 1766, at the age of 59. Although he became totally blind in 1771, half of his work was done during this second period in St. Petersbourg.

The most famous function in number theory:

$$\zeta(s) := 1^{-s} + 2^{-s} + 3^{-s} + \dots$$

made its first appearance in Euler's famous equation

$$1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \frac{1}{25} + \dots = \frac{\pi^2}{6}$$

in a letter to Daniel Bernoulli in 1736, [32]. Bernoulli found it "*sehr merkwürdig*", that is "very strange".

Euler extended his formula to

$$\zeta(2k) = 1^{-2k} + 2^{-2k} + 3^{-2k} + \ldots = \frac{(2\pi)^{2k}}{2(2k)!}\, B_k$$

where

$$\frac{x}{e^x - 1} = 1 - \frac{x}{2} + \sum_{k=1}^{\infty} (-1)^{k-1} B_k \, \frac{x^{2k}}{(2k)!}\,.$$

Euler tried hard to obtain a closed formula for $\zeta(3)$, to no avail. In his 1785 *Opuscula analytica* (see [27, T. II, §14, 91–101]) he writes:

"Huiusmodi igitur casum singularem, qui talem relationem non respuere videtur, hic accuratius evolvam, scilicet summam seriei reciprocæ cubarum $1 + \frac{1}{2^3} + \frac{1}{3^3} + \frac{1}{4^3} + \frac{1}{5^3} + \ldots$ *etc., quam nullo adhuc modo sive ad circulum sive ad logarithmos reducere potui."*

which can be translated as: "At this point I will examine in rather more detail a unique case, which does not seem alien to follow such a relation, namely the sum of the series of the reciprocals of cubes $1 + \frac{1}{2^3} + \frac{1}{3^3} + \frac{1}{4^3} + \frac{1}{5^3} + \ldots$ *etc.*, which so far in no way I could reduce to the circle or to logarithms."

He tried to look for linear relations with integer coefficients among π^3, $\pi^2 \log 2$, $(\log 2)^3$, almost certainly because of the identities

$$\log 2 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \ldots, \quad \frac{\pi^2}{6} = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \ldots,$$

hence the sum $\sum (-1)^n n^{-3}$ appears in the series obtained by multiplying the series for the factors of $(\log 2)^3$ or $\pi^2 \log 2$. He was able to show that no such relation can occur with coefficients below a certain numerical bound. This was not a trivial calculation to do by hand and he had to use a lot of ingenuity to do it.

The deviation from $\zeta(3)$ in such linear combinations is expressed by sums of the type

$$\sum_{m<n} m^{-a} n^{-b}$$

(or those with additional suitable signs ± 1 in the summands), the so-called multiple zeta sums. Thus he was led to study these new sums for many years, discovering several surprising identities among them.

The problem of understanding relations among multiple zeta sums remains of great interest, linked with algebraic geometry and the theory of Grothendieck's motives. A conjecture of Zagier [61] states that the space of $\mathbb{Q}$-linear relations among multiple zeta values

$$\zeta(a_1, a_2, \ldots, a_r) = \sum_{0<m_1<m_2<\cdots<m_n} m_1^{-a_1} m_2^{-a_2} \ldots m_r^{-a_r}$$

where $a_i \in \mathbb{N}$, $a_r \geq 2$, and $a_1 + a_2 + \cdots + a_r = n$, has for $n \geq 3$ dimension equal to the coefficient d_n of x^n in the power series expansion

$$\frac{1}{1 - x^2 - x^3} = \sum_{n=0}^{\infty} d_n x^n.$$

The first few values of d_n are $1, 1, 1, 2, 2, 3, 4, 5, 7, 9, 12, \ldots$. For example, the simplest relation is Euler's relation $\zeta(1,2) = \zeta(3)$, showing that $d_3 \geq 1$.

A proof that the dimension of this space does not exceed d_n was provided independently by the work of Terasoma [51] and Deligne and Goncharov [23], based on the theory of mixed Tate motives. The explicit relations proved by Euler and many others show indeed that Zagier's conjecture is true for low values of n.

In 1979, Apéry [2] proved the first result about the arithmetic structure of $\zeta(3)$, namely

Theorem (Apéry, 1979). *$\zeta(3)$ is irrational.*

This has been refined by several authors (Apéry, Beukers, Hata, Rhin and Viola, see [44]) to prove that $\zeta(3)$ is diophantine, meaning that there are positive constants A and κ such that

$$\left|\zeta(3) - \frac{p}{q}\right| > \frac{A}{q^\kappa}$$

for any rational fraction p, q. It is expected that any constant $\kappa > 2$ will do the job, but most likely this will require some major new idea; the present record value for κ is quite small, namely $\kappa = 5.513891$, see [44] and the bibliography therein.

Further results have been obtained about the values of $\zeta(2k+1)$. Rivoal [47] and Ball and Rivoal [3] have shown that the dimension of the $\mathbb{Q}$-linear space spanned by $k = 1, 2, \ldots, K$ goes to ∞ with order at least $\log K$. It is expected that these numbers are linearly independent over the rational field. Of course, the numbers $\zeta(2k)$ $(k = 1, 2, \ldots)$ are linearly independent over $\mathbb{Q}$, because of Euler's formula for $\zeta(2k)$ and Lindemann's theorem that π is transcendental.

In the first volume of Euler's book of 1748 *Introductio in Analysin Infinitorum* we find the formula

$$1^{-s} + 2^{-s} + 3^{-s} + \ldots = \prod \left(1 - \frac{1}{p^s}\right)^{-1}.$$

which is the famous Euler product. He used the fact that the series is divergent when $s = 1$ to deduce Euclid's theorem that the sequence of primes is infinite.[1]

Euler did not stop here with the zeta function. He conjectured the functional equation (see [29])

$$\frac{1 - 2^{n-1} + 3^{n-1} - 4^{n-1} + 5^{n-1} - 6^{n-1} \quad etc.}{1^{-n} - 2^{-n} + 3^{-n} - 4^{-n} + 5^{-n} - 6^{-n} \quad etc.} = \frac{-1 \cdot 2 \cdot 3 \cdots (n-1)(2^n - 1)}{(2^{n-1} - 1)\pi^n} \cos\frac{n\pi}{2}$$

but stated that, although it was almost certainly true, he could not prove it.

The functional equations for $L(s, \chi_{-3}) = 1 - \frac{1}{2^s} + \frac{1}{4^s} - \frac{1}{5^s} + \ldots$, $L(s, \chi_{-4}) = 1 - \frac{1}{3^s} + \frac{1}{5^s} - \ldots$ (also conjectured by Euler) and $L(s, \chi_{-8}) = 1 + \frac{1}{3^s} - \frac{1}{5^s} - \frac{1}{7^s} + \ldots$ were finally proved by Malmstén in his thesis of 1846 and published in 1849, see [42].

[1] Euclid studiously avoids the use of infinity in his work. His theorem is: "Given any number, there is a prime larger than that number."

Theorem (Malmstén, 1849).

$$L(1-s,\chi_{-3}) = \frac{2}{\sqrt{3}}\Big(\frac{2\pi}{3}\Big)^{-s}\sin\frac{\pi s}{2}\,\Gamma(s)L(s,\chi_{-3}),$$
$$L(1-s,\chi_{-4}) = \Big(\frac{\pi}{2}\Big)^{-s}\sin\frac{\pi s}{2}\,\Gamma(s)L(s,\chi_{-4}),$$
$$L(1-s,\chi_{-8}) = \frac{1}{\sqrt{2}}\Big(\frac{\pi}{4}\Big)^{-s}\sin\frac{\pi s}{2}\,\Gamma(s)L(s,\chi_{-8}).$$

In fact, Malmstén gives first a rigorous proof of the general functional equation

$$\sum_{n=0}^{\infty}\Big(\frac{1}{((2n+1)\pi-a)^s}-\frac{1}{((2n+1)\pi+a)^s}\Big) = \frac{1}{\sin(\frac{1}{2}\pi s)}\sum_{n=1}^{\infty}(-1)^{n-1}\frac{\sin(an)}{n^{1-s}}$$

and deduces the other three functional equations as special cases by setting $a=\pi/m$ with $m=2,\,3,\,4$. Malmstén's functional equation for $L(s,\chi_{-4})$ was given in the same year by Schlömilch [48] as an exercise for university students.

In April 1849, Eisenstein obtained a more general functional equation, giving a short sketch of proof (see [56]):

Theorem (Eisenstein, 1849). *For α, β in the open interval $(0,1)$ and for all s we have*

$$\sum_{n=1}^{\infty}\frac{e^{2\pi\alpha ni}}{(n-\beta)^{1-s}} = \frac{\Gamma(s)}{(2\pi)^s}\Big\{e^{s\frac{\pi i}{2}}\sum_{n=0}^{\infty}\frac{e^{2\pi\beta(n+\alpha)i}}{(n+\alpha)^s}+e^{-s\frac{\pi i}{2}}\sum_{n=1}^{\infty}\frac{e^{-2\pi\beta(n-\alpha)i}}{(n-\alpha)^s}\Big\}.$$

Lejeune Dirichlet's family came from the Belgian town of Richelet where Dirichlet's grandfather lived. This explains the origin of his name which comes from "Le jeune de Richelet" meaning "The young man from Richelet". His father was postmaster of Düren, the town of his birth situated about halfway between Aachen and Cologne. Even before he entered the Gymnasium in Bonn in 1817, at the age of 12, he had developed a passion for mathematics and spent his pocket-money on buying mathematics books. At the Gymnasium he was a model pupil.

He is recognized today for his fundamental contributions to number theory and analysis.

A *Dirichlet L-series* is given by

$$L(s,\chi) := \sum_{n=1}^{\infty}\frac{\chi(n)}{n^s}$$

where $\chi(n)$ is a *Dirichlet character* (mod q) with the following properties:

- $\chi(1)=1$ and $\chi(n)=0$ if n, q have a common factor.
- Multiplicativity: $\chi(mn)=\chi(m)\chi(n)$.
- Periodicity: $\chi(n+q)=\chi(n)$.

The period q is the *modulus* of the character. Such a character is *imprimitive* if there is another character χ^* to a modulus q^*, with q^* a proper divisor of q, with the property that $\chi(n) = \chi^*(n)$ whenever $(n, q) = 1$; it is called *primitive* otherwise. The *principal* character $\pmod q$, often denote by χ_0, is the character such that $\chi_0(n) = 1$ of $(n, q) = 1$ and is 0 otherwise. This character is imprimitive unless $q = 1$.

By multiplicativity and periodicity it is immediate that if χ is a character $\pmod q$ then $\chi^{q-1}(n)$ is 1 or 0 according as $(n, q) = 1$ or 0. The *order* of a character is the smallest positive integer m such that $\chi^m(n) = 1$ for $(n, q) = 1$. A character of order 2 is called a *quadratic* character.

The number of distinct characters to the modulus q is given by the Euler function $\phi(q)$.

The main property of characters to modulus q is embodied in the *orthogonality relations*:

$$\sum_{n=1}^{q} \chi_1(n)\overline{\chi_2}(n) = \phi(q) \quad \text{if} \quad \chi_1 = \chi_2$$

$$\sum_{\chi} \chi(m)\overline{\chi}(n) = \begin{cases} \phi(q) & \text{if} \quad m \equiv n \pmod q \text{ and } (n, q) = 1 \\ 0 & \text{otherwise} \end{cases}$$

These L-functions admit a Euler product

$$L(s, \chi) = \prod \left(1 - \frac{\chi(p)}{p^s}\right)^{-1}$$

and a functional equation (for χ *primitive*):

$$L(1 - s, \overline{\chi}) = \varepsilon_\chi 2^{1-s}\pi^{-s}q^{s-\frac{1}{2}} \begin{Bmatrix} \cos \frac{\pi s}{2} \\ \sin \frac{\pi s}{2} \end{Bmatrix} \Gamma(s)L(s, \chi)$$

with $|\varepsilon_\chi| = 1$ and the choice of cos or sin according to the sign of $\chi(-1) = \pm 1$.

A Dirichlet character $\pmod q$ is $\chi(n) = 0$ if $(n, q) > 1$ and for $(n, q) = 1$ is the pull-back to $\mathbb{Z}$ of a character of $(\mathbb{Z}/q\mathbb{Z})^*$. The orthogonality relations of characters of the group $(\mathbb{Z}/q\mathbb{Z})^*$ allow one to do harmonic analysis of arithmetic progressions $\pmod q$.

Theorem (Dirichlet, 1837). *Every arithmetic progression a, $a + q$, $a + 2q$, $a + 3q$, $a + 4q$, ... without fixed divisor contains infinitely many prime numbers.*

The proof is in three main steps:

- $$\sum_{p^m \equiv a \pmod q} \frac{1}{mp^{ms}} = \frac{1}{\phi(q)} \sum_{\chi} \overline{\chi}(a) \log L(s, \chi).$$
- $L(1, \chi) \neq 0$ *if χ is not the principal character.*
- *Let $s \to 1_+$ in the first equation.*

The first step uses orthogonality of characters and the Euler product, the formula holds if $s > 1$.

The crux of the proof is the second step. The difficult case is when χ is the non-trivial quadratic character. Dirichlet's proof in this case consists in a beautiful

identity expressing $L(1,\chi)$ in terms of the class number and fundamental unit of a certain quadratic field $\mathbb{Q}(\sqrt{D})$. The non-vanishing follows because, trivially, the class number is at least 1, hence not 0. The story is that Dirichlet, after several attempts, had the inspiration for his proof when listening to the music for Easter in the Sistine Chapel in Rome.

The third step is obvious: by the first and second steps, the right-hand side in the first formula diverges for $s \to 1+$, hence so is the left-hand side.

2. Approaching the Prime Number Theorem

Let $\pi(x)$ be the number of primes up to x. Euler wrote the equation

$$\sum \frac{1}{p} = \log\log\infty$$

as one finds at the very end of [30], p. 244 and [43], p. 11. He wrote (in slightly more modern notation)

$$A = \sum \frac{1}{p}, \quad B = \sum \frac{1}{p^2}, \quad C = \sum \frac{1}{p^3}, \quad \textit{etc.}$$
$$e^{A+\frac{1}{2}B+\frac{1}{3}C+\cdots} = \sum \frac{1}{n}$$
$$A = \log \sum \frac{1}{n} + \text{terms that can be neglected}$$

concluding with his formula. The foundation of this argument was probably based in the use of series, as in

$$1 + x + x^2 + x^3 + \cdots = \frac{1}{1-x}$$
$$1 + 1 + 1 + \cdots = \frac{1}{0} = \infty$$
$$x + \frac{1}{2}x^2 + \frac{1}{3}x^3 + \cdots = \log\frac{1}{1-x}$$
$$1 + \frac{1}{2} + \frac{1}{3} + \cdots = \log\frac{1}{0} = \log\infty,$$

ending with the logarithm of the formal Euler product:

$$\log\log\infty = \log\sum\frac{1}{n} = \log\prod\frac{p}{p-1} = A + \frac{1}{2}B + \frac{1}{3}C + \ldots.$$

Later, in 1762/63 Euler wrote (see [43] and [31], pp. 101–102):

"Ac primo quidem nullum est dubium, quin proposito numero quantumvis magno infiniti adeo existant numeri primi illo maiores, postquam iam ab Euclide est demonstratum omnium numerorum primorum multitudinem esse infinitam, etiamsi, ut ego ostendi, haec numerorum primorum multitudo se habeat ad multitudinem omnium prorsus numerorum, ut unitas ad infinitum seu potius ut logarithmus numeri infiniti ad ipsum hunc numerum infinitum, quod posterius infinitum maius est quam potestas quantumvis magna illius infiniti."

that may be translated as:[2] "The first point is that there is no doubt that, given an arbitrarily large number there are infinitely many prime numbers greater than that number, because it has been already proved by Euclid that the totality of prime numbers is infinite, even if, as I proved, the totality of prime numbers compares with the totality of all numbers as unity compares with infinity or, better, as the logarithm of the infinity of all numbers compares with the same infinity, which infinity, as a second point, is larger than any power of that infinity."[3]

In this writing Euler sets a multiple proportion $a : b = c : d = e : f$ where a is the totality of all prime numbers, b the totality of all natural integers, $c = 1$, $d = \infty$, $e = \log \infty$, $f = \infty$. Thus at first sight it appears that here Euler asserts first that $\pi(x) = o(x)$ and then that $\pi(x)$ has order of magnitude $\log x$. This is also found in the same letter to Goldbach cited in the footnote, where he asserts that "the number of primes is an infinity of the smallest order" compared with the infinity of numbers, again mentioning $\log n$ as an approximation for the number of primes up to n. This, although hard to believe as written (Euler had studied a table of prime numbers up to 101000), may be explained by Euler's approximation for the harmonic series of primes up to n compared with the harmonic series of the natural integers up to n, because the former is proportional only to $\log \log n$ and the latter is proportional to $\log n$, a fact that he discovered and published in the earlier paper [30].

The correct approximation $\pi(x) \sim \mathrm{li}(x)$, where $\mathrm{li}(x)$ is the integral logarithm function

$$\mathrm{li}(x) = \int_0^x \frac{\mathrm{d}t}{\log t}$$

was found by Gauss around 1792, when he was a teenager.

Legendre in 1796 observed that

$$\pi(x) \sim \frac{x}{\log x - 1.08366}$$

is a good approximation in the known range of numerical tables.

The approximation $\pi(x) \sim \mathrm{li}(x)$ is mentioned in a letter of Bessel to the astronomer Olbers in 1810, see [43], p. 97. There Bessel credited Gauss with $\pi(400000) = 33859$ and $\mathrm{li}(400000) = 33922.621995$, which makes a difference of less than 0.2%. (Actually, $\pi(400000) = 33860$ and $\mathrm{li}(400000) = 33922.621935\ldots$, but perhaps a badly handwritten 3 may look like a 9.) It is likely that by 1810 the Gauss approximation to $\pi(x)$ was widely known.

Dirichlet stated, in 1838 (Crelle, Bd.18 (1838), p. 272):

"*J'ai fait l'application des ces principes à la démonstration de la formule remarquable que Legendre a donnée pour exprimer d'une manière très approchée combien il y a de nombres premiers au-dessous d'une limite quelconque*".

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[2] Many thanks to my sister Professor Maria Giovanna Bombieri for help with the translation of this difficult Latin passage.

[3] Euler refers here to a comparison of infinities, in the form $\infty > (\log \infty)^m$, for any positive integer m. The fact that $\log n$ is smaller than any positive power of n is mentioned in a letter to Goldbach of 1752, see [32], p. 587; most likely, Euler noticed this from the exponential series.

The "*principes*" referred to by Dirichlet in this paper consist in the systematic use of Lambert series

$$\sum a_n \frac{x^n}{1-x^n}.$$

Interestingly enough, a footnote to this statement in Dirichlet's *Werke*, Bd. 1, Berlin 1889, p. 372, continues "*qui n'est exacte que dans son premier terme, la véritable expression limite étant* $\sum \frac{1}{\log n}$", with a footnote to this footnote [by the editor Kronecker] stating "*Anmerkung von Lejeune Dirichlet's Hand in dem an Gauss geschickten Exemplar*", namely "Annotation in Lejeune Dirichlet's handwriting in the offprint belonging to Gauss".

The first rigorous result on the distribution of prime numbers was obtained in 1848 by Čebyshev, who proved that if $\pi(x)$ admitted an asymptotic expansion in decreasing powers of a logarithm, then this expansion had to agree with the Gauss approximation. In this paper, by exploiting the Euler product, he wrote the logarithm of the zeta function as an integral transform involving $\pi(x)$ and deduced his result from his formula. However, he was unable to do the inversion the formula so to obtain an expression for $\pi(x)$ in terms of the zeta function. Čebyshev's result is:

Theorem (Čebyshev, 1848). *If $\pi(x)$ has a simple approximation, then*

$$\pi(x) \sim \int^x \frac{\mathrm{d}t}{\log t} \sim \frac{x}{\log x} + \frac{1!x}{(\log x)^2} + \frac{2!x}{(\log x)^3} + \dots$$

Gauss, in a famous letter to Hencke written on Christmas Eve (1849), wrote the approximation

$$\#\{p : a < a < p \le b\} \sim \int_a^b \frac{\mathrm{d}n}{\log n},$$

mentioning that he figured out this approximation already in 1792 or 1793 and that he scribbled it out on the back of a copy of a table of logarithms. Apparently, he was answering a suggestion by the astronomer Hencke that the constant in Legendre's formula might be $1/(2\log_{10} e) = 1.1513\dots$. In astronomy, all "natural" logarithms were calculated in base10, not in base e!

Ten years had to go by before Riemann solved the problem of the inversion of Čebyshev's formula when he wrote his epoch making memoir of 1859 on the number of primes below a given bound, showing that this required the complex Fourier transform (i.e the Mellin transform).

3. Riemann

Riemann's mathematical career started in Göttingen, but after two years he left it for Berlin in 1847, where he remained two years, mainly under the influence of Dirichlet. Returning to Göttingen, where Gauss was the leading star, he spent five hard years, with very little income, getting his doctorate in 1851 with a landmark

thesis on complex function theory. In 1854 he wrote his *Habilitation* essay on the foundations of geometry, namely the study of curved space of arbitrary dimension.

In 1859 Riemann [45] wrote his only paper in number theory, where he laid the foundations of the theory of the zeta function as a function of one complex variable and where he was able to exploit Euler's product formula so to obtain his explicit exact formula for the number of primes up to a given bound.

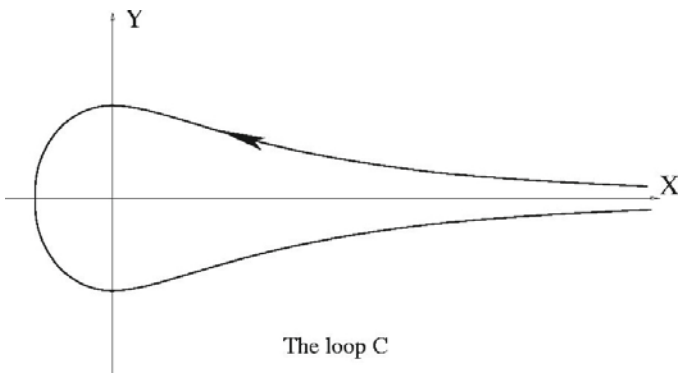
Riemann starts with the formula[4]

$$\Gamma(s)\zeta(s) = \int_0^\infty \frac{x^{s-1}}{e^x - 1}\,\mathrm{d}x$$

valid for[5] $\Re(s) > 1$ and deduces, after doubling the contour and deforming it, the new formula

$$2\sin(\pi s)\Gamma(s)\zeta(s) = i\int_L \frac{(-z)^{s-1}\mathrm{d}z}{e^z - 1}$$

where L is the loop



The loop C

encircling the origin and leaving outside the points $z = 2\pi ki$ for $k \pm 1, \pm 2, \ldots$, namely the other poles of the integrand (the branch of $\log(-z)$ for defining $(-z)^{s-1}$ is chosen so that it is real for negative z). The formula holds for all complex s.

Deforming the loop to the left and applying the calculus of residues yields

$$\zeta(1-s) = 2(2\pi)^{-s}\cos\Big(\frac{\pi s}{2}\Big)\,\Gamma(s)\zeta(s).$$

This method of Riemann also proves quite easily Eisenstein's functional equation.

Actually, Riemann stops here by writing the functional equation in the form

$$2\sin(\pi s)\Gamma(s)\zeta(s) = (2\pi)^s \sum n^{s-1}[(-i)^{s-1} + i^{s-1}]$$

and deduces, by the known duplication formula of the Gamma function, the second, more familiar, form of the functional equation:

$$\pi^{-\frac{s}{2}}\Gamma\Big(\frac{s}{2}\Big)\zeta(s) = \pi^{-\frac{1-s}{2}}\Gamma\Big(\frac{1-s}{2}\Big)\zeta(1-s).$$

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[4] The formula $\int_0^\infty t^{2n-1}\mathrm{d}t/(e^t - 1) = (2n-1)!(1 + 2^{-2n} + 3^{-2n} + \ldots)$ already occurs in the early work of Abel [1], T.I, p. 22.

[5] By $\Re(z)$ and $\Im(z)$ we denote the real and imaginary part of the complex number z.

Riemann then continues by giving a direct proof of the functional equation in this form by expressing one side as the Mellin transform of a theta function and using the known Jacobi functional equation for the theta function.

Riemann's formula is expressed as follows.

Riemann's Theorem, I. *If $x > 1$ is not a prime power, then*

$$f(x) := \sum_{n=1}^{\infty} \frac{1}{n}\pi(x^{1/n}) = \mathrm{li}(x) - \sum_{\rho} \left\{\mathrm{li}(x^{\rho}) + \mathrm{li}(x^{1-\rho})\right\} + \int_x^{\infty} \frac{\mathrm{d}t}{(t^2-1)t\log t} - \log 2,$$

where ρ runs over the non-trivial zeros of $\zeta(s)$ and

$$\mathrm{li}(x^{\beta+i\gamma}) = \int_{-\infty+i\gamma\log x}^{(\beta+i\gamma)\log x} \frac{e^z \mathrm{d}z}{z + \mathrm{sign}(\gamma)\pi i}.$$

If $x = p^k$ is a prime power, then the theorem still holds provided we correct the definition of $f(x)$ by adding the quantity $1/(2k)$.

Riemann's Theorem, II.

$$\pi(x) = \sum_{n=1}^{\infty} \frac{\mu(n)}{n} f(x^{1/n})$$

where $f(x)$ is the left-hand side of Riemann's Theorem and $\mu(n)$ is the Möbius function.

Riemann gave only a formal proof with the calculus of residues involving the function $\log\zeta(s)$. The constant $-\log 2$ was given incorrectly as $\log\xi(0)$.[6]

In fact, Riemann's notebook shows that he also considered the logarithmic derivative

$$-\frac{\zeta'}{\zeta}(s) = \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^s}$$

which is meromorphic in the whole complex plane. He wrote there the correct formula

$$\psi(x) := \sum_{n\le x} \Lambda(n) = x - \sum \frac{x^{\rho}}{\rho} - \frac{1}{2}\log(1-x^{-2}) - \log(2\pi)$$

valid for $x > 1$ not an integer (with the same grouping of terms in the sum as in his published formula).

Already in his memoir Riemann saw the fundamental role played by the the zeta function, as a function of a complex variable, in the study of prime numbers.

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[6] This was pointed out by Genocchi already back in 1860, see [43], p. 145 and the reference therein. In the French edition of his collected papers [45], p. 176, it is suggested, probably in deference to Riemann: "It is very likely that this be nothing else than a "*lapsus calami*", or misprint, namely $\log\xi(0)$ for $\log\zeta(0)$; in fact, $\zeta(0) = \frac{1}{2}$." However, $\zeta(0) = -\frac{1}{2}$; rather, the original manuscript by Riemann[46] shows a clearly written $\log\xi(0)$ throughout.

In his memoir, Riemann formulates the celebrated

Riemann hypothesis. *All complex zeros of* $\zeta(s)$ *have real part* $\frac{1}{2}$.

He actually states that the number of zeros of $\xi(t)$ with real part up to height T is $\frac{T}{2\pi}\log\frac{T}{2\pi} - \frac{T}{2\pi}$ up to a percentage fraction of order $\frac{1}{T}$ and continues: "Indeed, one finds within these limits a number of real zeros of $\xi(t)$ approximately equal to this number and it is very likely that all roots of $\xi(t)$ are real." He concludes: "One would like to have an exact proof of that; however after a few hasty unsuccessful trials I put aside this search for the time being, since this did not seem to be necessary for the current purposes of my research."

4. The classical theory of the zeta function

In this section we review many results about the analytic behavior of the zeta function. The interested reader will find further references and proofs in the standard monograph by Titchmarsh, [52] (second edition, annotated by D.H. Heath-Brown).

From Riemann's first formula for $\zeta(s)$, the functional equation, and the Euler product, one gets

- $\zeta(s) \neq 0$ if $\Re(s) > 1$.
- $\zeta(s)$ has a simple pole at $s = 1$ and is analytic for $s \neq 0$.
- $\zeta(s) = 0$ for $s = -2, -4, -6, \ldots$ (the *trivial* zeros).
- $\zeta(s)$ has polynomial growth in $|s|$ in every fixed vertical strip in the s-plane (excluding of course a neighborhood of the pole at $s = 1$).

A method of proof for the last step which extends to other situations:

- $\zeta(s)$ is of polynomial growth in any strip sufficiently to the right (absolute convergence) and to the left (functional equation).
- $\zeta(s)$ is meromorphic of finite order with finitely many poles, so the Phragmén–Lindelöf Theorem applies here.

Let $s = \sigma + it$ and define

$$\mu(\sigma) := \limsup_{|t|\to\infty} \frac{\log|\zeta(\sigma + it)|}{\log|t|}.$$

Then the Phragmén–Lindelöf theorem shows that

$$\mu(\sigma) \quad \textit{is a convex function of } \sigma$$

and the functional equation shows that

$$\mu(1-\sigma) = \sigma - \frac{1}{2} + \mu(\sigma).$$

It is clear that $\mu(\sigma) = 0$ for $\sigma > 1$, hence by convexity

$$\mu(\sigma) \leq \frac{1}{2}(1-\sigma) \quad (0 \leq \sigma \leq 1).$$

The Lindelöf hypothesis is

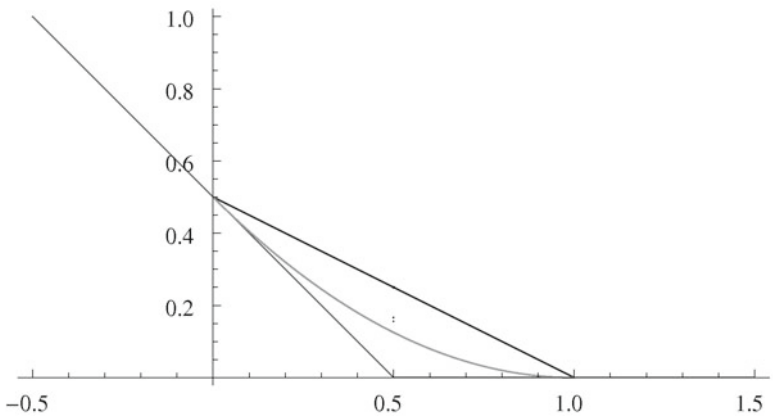
$$\mu\big(\frac{1}{2}\big) = 0.$$

The Lindelóf hypothesis has many important arithmetic consequences. It is equivalent to the statement that if $N(\alpha, T)$ is the number of zeros of $\zeta(s)$ in the rectangle $\alpha \le \sigma \le 1$, $0 \le t \le T$, then for any fixed $\alpha > \frac{1}{2}$ we have

$$N(\alpha, T+1) - N(\alpha, T) = o(\log T).$$

For arithmetic applications, it is often sufficient to show that the convexity bound is never sharp in $0 < \sigma < 1$, in other words $\mu(\frac{1}{2}) < \frac{1}{4}$. This is done by approximating the zeta function by a truncation of the Dirichlet series and using non-trivial bounds for the associated trigonometrical sums. Some bounds that have been obtained in succession:

$$\mu\big(\frac{1}{2}\big) \le \frac{1}{6}, \frac{163}{988}, \dots, \frac{15}{92}, \dots, \frac{9}{56}, \dots, \frac{32}{205}, \dots$$



The Lindelöf μ-function

In the picture: The convexity interpolation between 0 and 1; the point $(\frac{1}{2}, \frac{1}{6})$ and the best known upper bound for $\mu(\frac{1}{2})$, namely the point $(\frac{1}{2}, \frac{32}{205})$; the parabola $(1-\sigma)^2/2$; the graph of $\mu(\sigma)$ on the Lindelöf hypothesis.

Theorem. *Let $N(T)$ be the number of zeros $\rho = \beta + i\gamma$ of $\zeta(s)$ with $0 < \gamma \le T$, T not the imaginary part of a zero. Then*

$$N(T) = \frac{T}{2\pi} \log \frac{T}{2\pi} - \frac{T}{2\pi} + \frac{7}{8} + S(T) + O\big(\frac{1}{T}\big)$$

complex zeros with $0 \le \Im(s) \le T$, where $S(T) = \frac{1}{\pi} \arg \zeta\big(\frac{1}{2} + iT\big)$ with the argument taken along the half-line $(\infty + iT, \frac{1}{2} + iT)$, starting with the value 0 at $\infty + iT$.

This was essentially stated by Riemann and proved by von Mangoldt [53] in 1895 with the error term $S(T) = O((\log T)^2)$. Later, Backlund (1918) improved the error term to $S(T) = O(\log T)$, still the best known bound.

Theorem (Littlewood, 1924). *On the Riemann hypothesis,*

$$\log \zeta\Big(\frac{1}{2} + it\Big) \ll \frac{\log T}{\log \log T}, \quad \int_0^T S(t)\, \mathrm{d}t \ll \frac{\log T}{(\log \log T)^2}$$

and $\ll \log T$ unconditionally.

Bohr and Landau in 1914 proved the bound $N(\alpha, T) \ll T$ for any fixed $\alpha > \frac{1}{2}$, showing that the zeros of $\zeta(s)$ tend to cluster along the critical line $\sigma = \frac{1}{2}$ and a few months later in [6] they improved this to $N(\alpha, T) = o(T)$, also for Dirichlet L-functions.

This improvement, albeit very small, turns out to be significant because, as we shall see in Section 15, there are Dirichlet series, satisfying a functional equation of the same type as for the zeta function, for which $N(\alpha, T) > c(\alpha)T$ for some $c(\alpha) > 0$ and all large T, whenever $\frac{1}{2} < \alpha < \alpha^*$ with a certain $\alpha^* > 1$. The important new idea introduced in this paper was to multiply the zeta or L-function by a truncation of the reciprocal of the Euler product, thus reducing drastically the main term in the L^2-mean-value used to estimate the number of zeros off the line. This was the first example of what today is called a *mollifier*.

A major step forward was done by Carlson [14] in 1920, who proved the much stronger result

$$N(\alpha, T) \ll T^{1-(2\alpha-1)^2}(\log T)^5;$$

in particular, the exponent of T is less than 1. Carlson's further idea was to use as a mollifier a truncation of the inverse of the zeta function, rather than a truncation of the reciprocal of the Euler product. The success of the method depends in a crucial way on the fact that the coefficients of the Dirichlet series

$$\frac{1}{\zeta(s)} = \sum_{n=1}^{\infty} \frac{\mu(n)}{n^s},$$

are bounded by $O(n^{\varepsilon})$ for any $\varepsilon > 0$, in fact $|\mu(n)| \leq 1$ in this special case.

A quantitative form of this is expressed by

Theorem. *Let $f(s) = \sum a_n n^{-s}$ be a Dirichlet series with $a_1 = 1$, $a_n = O(n^{\varepsilon})$ for any fixed $\varepsilon > 0$. Define the coefficients b_n by*

$$\frac{1}{f(s)} = \sum_{n=1}^{\infty} \frac{b_n}{n^s}$$

and assume that $f(s)$ has a zero ρ with $\Re(\rho) = \alpha > 1$. Then we have

$$\max_{d<X} |b_d| > X^{\alpha-\frac{1}{2}-\delta}$$

for any fixed $\delta > 0$ and all sufficiently large X.

We leave the proof of this result (and other variants and refinements) to the interested reader, as an exercise.

Carlson result, together with a wide zero-free region for $\zeta(s)$ proved by Littlewood using the theory of Weyl sums, turned out to be crucial in Hoheisel's proof that the gap between consecutive primes p_n and p_{n+1} was $p_{n+1} - p_n \ll p_n^{1-\delta}$ for some positive δ and, more precisely, for obtaining a Prime Number Theorem in short intervals.

Let $c(\alpha)$ be the best exponent for which $N(\alpha, T) \ll T^{c(\alpha)+o(1)}$. Linear interpolation would give

Density Hypothesis. $c(\alpha) \le 2(1-\alpha)$.

The density hypothesis can be used as a substitute for the Riemann hypothesis in several applications, for example in obtaining a Prime Number Theorem in a short interval $(x, x + x^{\frac{1}{2}+\varepsilon})$, thus much work has been done towards better density theorems. The standard technique is as follows.

- The number of zeros in a bounded domain D is $1/(2\pi)$ times the variation of the argument of $\log \zeta(s)$ along the boundary ∂D of the domain.[7]
- A majorant for the number of zeros is also obtained if we replace $\zeta(s)$ by $f(s)\zeta(s)$ with $f(s)$ holomorphic in D. Then replace $\zeta(s)$ by a Dirichlet polynomial approximation and take for $f(s)$ a Dirichlet polynomial approximation for $\zeta(s)^{-1} = \sum \mu(n) n^{-s}$ (note that $|\mu(n)| \le 1$).
- Use convexity estimates and averaging techniques to replace log by L^p norms.

It is known that the Lindelöf hypothesis implies the density hypothesis for $\frac{1}{2} < \alpha < \frac{3}{4}$ and even $c(\alpha) = 0$ for $\alpha \ge \frac{3}{4}$ (Hálasz and Turàn, 1969). The method of Hálasz and Turàn suffices to prove the density hypothesis, and even more, in a neighborhood of $\alpha = 1$. This method went through several refinements with Jutila, Heath-Brown, and Bourgain. Today it is known that the density hypothesis holds for $\frac{25}{32} = 0.78125 < \alpha \le 1$, see [12]. Remarkably, the bound $c(\alpha) \le 3(1-\alpha)/(2-\alpha)$ proved by Ingham in 1940 remains the best known bound in the range $\frac{1}{2} < \alpha < \frac{3}{4}$.

The problem of the growth of $\zeta(s)$ in the critical strip is very important, especially for s on the critical line. The study of moments has been successful for moments of order 2 and 4. It has been proved that

$$\int_0^T \left|\zeta\Big(\frac{1}{2}+it\Big)\right|^2 \mathrm{d}t = TP_1(\log T) + O(T^{\theta_1+\varepsilon})$$

$$\int_0^T \left|\zeta\Big(\frac{1}{2}+it\Big)\right|^4 \mathrm{d}t = TP_2(\log T) + O(T^{\theta_2+\varepsilon})$$

and it has been conjectured that for all positive integers k

$$\int_0^T \left|\zeta\Big(\frac{1}{2}+it\Big)\right|^{2k} \mathrm{d}t = TP_k(\log T) + O(T^{\theta_k+\varepsilon})$$

where $P_k(x)$ is a polynomial of degree k^2 and $\theta_k < 1$.

The moment conjecture implies the Lindelöf hypothesis. Dirichlet polynomials $\sum a_n n^{-it}$ are not efficient for approximating purposes when $n \gg t$ because then n^{-it} does not oscillate when going from n to $n+1$, hence special features of the coefficients a_n are hard to exploit.

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[7] Some smoothness condition on the boundary is assumed, for example that it is piece-wise smooth. Zeros at a smooth point of the boundary are counted with half-multiplicity and at a corner with $1/(2\pi)$ the angle of the corner times the multiplicity.

For the zeta function, one can obtain excellent approximations with Dirichlet polynomials of length about $\sqrt{|t|/(2\pi)}$, sufficient for moments of order 2 and 4. We have (Hardy and Littlewood (1918) and Ingham (1926)):

$$P_1(x) = x + (2C - 1 - \log(2\pi)), \quad P_2(x) = \frac{1}{2\pi^2}x^4 + \dots.$$

The problem of conjecturing an exact formula for the first coefficient of the polynomial $P_k(x)$ turns out to be highly nontrivial. Conrey and Ghosh [19] (1989) and Conrey and Gonek [20] (2001) used methods of analytic number theory to conjecture

$$P_3(x) = \frac{42}{9!}a(3)x^9 + \dots, \quad P_4(x) = \frac{24024}{16!}a(4)x^{16} + \dots.$$

where $a(\lambda)$ is a certain arithmetically defined function of λ.

The arithmetic factor $a(\lambda)$ was encountered by Titchmarsh in 1928 and is given by

$$a(\lambda) = \prod_p \Big(1 - \frac{1}{p}\Big)^{\lambda^2} \times \Big(\sum_{n=1}^{\infty} \frac{d_\lambda(n)^2}{n^2}\Big)$$

where $\zeta^\lambda(s) = \sum d_\lambda(n)n^{-s}$ (with the branch $1^\lambda = 1$).

This factor $a(\lambda)$ arises by approximating the zeta function with a short Dirichlet polynomial; it yields a lower bound for the moment, but without the numerical factor in front of it. If we write this numerical factor as $g(\lambda)/\Gamma(\lambda^2+1)^2$ then the question is what should be $g(\lambda)$. Then $g(1) = 1$, $g(2) = 2$, $g(3) = 24$, $g(4) = 24024, \dots$ Then in 2000 Keating and Snaith [38] showed that

$$g(\lambda)/\Gamma(\lambda^2+1)^2 = G^2(\lambda+1)/G(2\lambda+1)$$

corresponds to the distribution of moments of the characteristic polynomial of random unitary matrices (the GUE ensemble); here $G(z)$ is the Barnes G-function ($G(z+1) = \Gamma(z)G(z)$, $G(1) = 1$) namely the natural interpolation of the running factorial, so

$$G(n+1) = 1!\,2!\,3!\cdots n!.$$

Thus the next conjecture is $g(5) = 701149020$.

When $N \sim \frac{1}{2\pi}\log T$, the GUE distribution is what is expected for the renormalized $t(\gamma) = \frac{1}{2\pi}\gamma\log\gamma$ where $\frac{1}{2} + i\gamma$ runs over zeros of the Riemann zeta function (Montgomery, 1973). A sketch of the beautiful idea of Keating and Snaith, in very condensed form, is as follows.

- When $N \sim \frac{1}{2\pi}\log T$, eigenangles t_n of $N \times N$ unitary matrices A correspond to $t(\gamma)$ for $0 < \gamma < T$, *via* rescaling.
- The Haar measure on $U(N)$ is

$$\mathrm{d}\mu_N = (2\pi)^{-N}(N!)^{-1}\prod_{m<n}\big|e^{it_m} - e^{it_n}\big|^2\mathrm{d}t_1\cdots\mathrm{d}t_N.$$

- Thus if $N \sim \frac{1}{2\pi}\log T$ the moment of order λ should be related to

$$\int_{U(N)}\big|\det(I - Ae^{-it})\big|^\lambda\mathrm{d}\mu_N = \int_{U(N)}\prod\big|e^{it} - e^{it_n}\big|^\lambda\mathrm{d}\mu_N.$$

- The integral is a special case of the Selberg integral.

This fundamental idea has led to extremely interesting developments.

5. The distribution of values of $\log\zeta\left(\frac{1}{2}+it\right)$

The distribution of values of $\log\zeta(\frac{1}{2}+it)$ has been determined (unconditionally!) by Selberg.

Theorem (Selberg, 1946). *The functions*

$$\frac{\log\left|\zeta\left(\frac{1}{2}+it\right)\right|}{\sqrt{\pi\log\log|t|}},\quad \frac{\arg\zeta\left(\frac{1}{2}+it\right)}{\sqrt{\pi\log\log|t|}},$$

become distributed, in the limit of large t, like independent random variables having Gaussian density $\exp(-\pi u^2)\mathrm{d}u$. *Hence for any bounded domain B with piece-wise smooth boundary we have*

$$\lim_{T\to\infty}\frac{1}{T}\operatorname{meas}\left\{t\in[T,2T]:\left(\frac{\log\left|\zeta\left(\frac{1}{2}+it\right)\right|}{\sqrt{\pi\log\log t}},\frac{\arg\zeta\left(\frac{1}{2}+it\right)}{\sqrt{\pi\log\log t}}\right)\in B\right\}$$
$$=\int_B e^{-\pi(x^2+y^2)}\mathrm{d}x\mathrm{d}y.$$

The proof depends on obtaining asymptotics for the moments. It goes through the following steps.

- Obtain approximations for $\log\zeta\left(\frac{1}{2}+it\right)$ by means of short Dirichlet polynomials Σ_X of length X and a correction U_X determined by the zeros of $\zeta(s)$ with $\beta\geq\frac{1}{2}$. A delicate density theorem $N(\alpha,T)\ll T^{1-\eta(\alpha-\frac{1}{2})}\log T$ with only one logarithm (a higher power will not do) is essential for handling the error terms introduced by the correction U_X.
- For $X<T^{\varepsilon_0/(a+b)}$, compute asymptotics for the moments:

$$\int_0^T(\Sigma_X)^a(\overline{\Sigma_X})^b\mathrm{d}t=(\delta(a-b)a!b!+o(1))\,T(\log\log X)^{a+b}.$$

- Prove an L^1-estimate

$$\int_0^T\left|\log\zeta\left(\frac{1}{2}+it\right)-\Sigma_X\right|\mathrm{d}t=O\left(T\frac{\log T}{\log X}\right).$$

- Appeal to standard results of probability theory to conclude.

An extension and a simplified proof of Selberg's theorem to $GL(2)$ L-functions (but without the error term in the asymptotics) is given in [8].

6. Zeros on the critical line

Hardy (1914) was the first to prove that $\zeta(s)$ has infinitely many zeros on the critical line. Subsequent methods proved only that the number $N_0(T)$ of zeros on the critical line in $(0,T)$ was $N_0(T) \gg T$.

Finally, Selberg (1941) succeeded in proving $N_0(T) \gg T \log T$, by working not with $\zeta(s)$ directly but rather with $\psi_X(s)\zeta(s)$ where $\psi_X(s)$ is a well-chosen Dirichlet polynomial approximation to $1/\zeta(s)$ (i.e. a mollifier) of length X. Here $X = T^c$, with c any positive constant not exceeding a suitable bound; the constant involved in the lower bound for $N_0(T)$ goes to 0 with c.

Selberg's method also extends to Epstein zeta functions associated to binary positive definite quadratic forms, namely

$$E(z,s) = \sum_{(m,n)\neq(0,0)} \frac{\Im(z)^s}{|mz+n|^{2s}}$$

where $\Im(z) > 0$. These functions satisfy the functional equation

$$\pi^{-s}\Gamma(s)E(z,s) = \pi^{-(1-s)}\Gamma(1-s)E(z,1-s),$$

but in general do not satisfy a Riemann hypothesis. In fact, except possibly for a few cases, the number of zeros off the critical line up to height T in the half-plane $\alpha < \Re(s)$ with $\alpha > \frac{1}{2}$ is $\sim c(\alpha)T$ with

$$c(\alpha) = \int_\alpha^\infty \mu(\tau)\mathrm{d}\tau$$

for a certain positive density $\mu(\tau)$. The function $\mu(\tau)$ admits right- and left-derivatives at every point and is not constant. For this, see the papers by Bohr and Jessen [5], Borchsenius and Jessen [11], and Yoonbok Lee [39].

The expected few cases when a Riemann hypothesis might hold for an Epstein zeta function can occur only when z is algebraic of degree 2 and $\mathbb{Q}(z)$ has class number 1 and otherwise the Epstein zeta function has infinitely many zeros with real part greater than 1. This was proved by Davenport and Heilbronn in [21], [22], when the quadratic form $|mz+n|^2$ is a rational form. If the form is transcendental the method of [21] also proves the same result for an Epstein zeta function. The much more delicate case when the quadratic form is irrational with algebraic coefficients can also be treated with the clever method used by Cassels [15] to prove the analogous result for the Hurwitz zeta function.

Another method, due to Levinson [41] in 1974 combines a different approach by Siegel with the mollifier technique of Selberg. This method requires the computation of several delicate asymptotics and one emerges with the correct lower bound $N_0(T) \gg T \log T$ only at the end. Levinson's method is better suited than Selberg's for numerical results and shows that more than 34% of zeros of $\zeta(s)$ lie on the critical line and are simple. On the other hand, Selberg's method is more flexible and can be applied with success to other functions, such as Epstein zeta functions and $GL(2)$ L-functions.

This has been subsequently improved by Conrey [18] in 1989 to

$$N_0(T) > 0.4077\, N(T)$$

for large T. Conrey's method uses deep results to deal with mollifiers ψ_X with $X = T^{4/7-\varepsilon}$, while early methods stopped with shorter mollifiers with $X \asymp T^{1/2}$. Small improvements on this have been obtained by other authors with the use of slightly longer mollifiers.

Numerical computations have been used to show that the Riemann hypothesis holds for $0 < t < T_0$ for suitable values of T_0. Riemann's *Nachlass* contains a numerical calculation of the first few zeros. Apparently, it has gone unnoticed that it also contains the calculation of the first few solutions of $\Re\big(\zeta\big(\frac{1}{2}+it\big)\big) = 0$.

The largest T_0 computed by hand is $T_0 = \gamma_{1041}$ (Titchmarsh and Comrie, 1936).

The current verification using a computer grid shows that the first 10^{13} zeros satisfy the Riemann hypothesis, corresponding to

$$T_0 \asymp 2.61442 \times 10^{11},$$

(Gourdon and Demichel, 2004). Higher batches up to 10^{20} were computed by Odlyzko (1992) for the purpose of checking the validity of the GUE distribution in these high ranges, with excellent agreement.

A chronology of computations of zeros of the zeta function:

Riemann (around 1859) computed at least the first three zeros of the zeta function by the Riemann–Siegel formula.

von Mangoldt (1894) showed that the first complex zero of the zeta function has real part $\frac{1}{2}$.

De La Vallée-Poussin (1896) showed that the first three complex zeros of the zeta function have real part $\frac{1}{2}$.

Gram (1903) used Euler-Maclaurin summation and discovered Gram's law. He showed that all 10 zeros with $\Im(\rho) \le 50$ range lie on the critical line with real part $\frac{1}{2}$, by computing the sum of the inverse 10th powers of the roots he found.

Backlund (1914) introduced a better method of checking that all the zeros up to $\Im(\rho) \le 200$ on the line, by studying the argument $S(T)$ of the zeta function.

Hutchinson (1925) extended found the first failure of Gram's law, at the Gram point g_{126}.

1935, $n = 195$. Titchmarsh used the recently rediscovered Riemann-Siegel formula, which is much faster than Euler-Maclaurin summation. It takes about $O(T^{3/2+\varepsilon})$ steps to check zeros with imaginary part less than T, while the Euler-Maclaurin method takes about $O(T^{2+\varepsilon})$ steps.

1936, $n = 1041$. Titchmarsh and Comrie were the last mathematicians to compute zeros by hand.

1953, $n = 1104$. Turing found a more efficient way to check that all zeros up to some point are accounted for by the zeros on the line, by checking that $Z(t)$ has the correct sign at several consecutive Gram points and using the fact that $S(T)$ has average value 0. This requires almost no extra work because the sign of $Z(t)$ at Gram points is already known from finding the zeros, and is still the usual method used. This was the first use of a digital computer to calculate the zeros.

1956, $n = 15,000$. Lehmer discovered a few cases where the the zeta function has zeros that are "only just" on the line: two zeros of the zeta function are so close together that it is unusually difficult to find a sign change between them. This is called the "Lehmer's phenomenon", and first occurs at the zeros with imaginary parts 7005.063 and 7005.101, which differ by only .04 while the average gap between other zeros near this point is about 1.

1956, $n = 25,000$. Lehmer.

1958, $n = 35,337$. Meller.

1966, $n = 250,000$. Lehman.

1968, $n = 3,500,000$. Rosser, Yohe & Schoenfeld stated Rosser's rule.

1977, $n = 40,000,000$. Brent.

1979, $n = 81,000,001$. Brent.

1982, $n = 200,000,001$. Brent, van de Lune, te Riele, Winter.

1983, $n = 300,000,001$. van de Lune, te Riele.

1986, $n = 1,500,000,001$. van de Lune, te Riele, and Winter gave some statistical data about the zeros and gave several graphs of $Z(t)$ at places where it has unusual behavior.

1987. A few of large height: Odlyzko computed smaller numbers of zeros of much larger height, around 10^{12}, to high precision to check Montgomery's pair correlation conjecture.

1992, A few of large height : Odlyzko computed a few zeros of heights up to 10^{20}, and gave an extensive discussion of the the results.

2001, $n = 10,000,000,000$. van de Lune (unpublished)

2004, $n = 900,000,000,000$. Wedeniwski (ZetaGrid distributed computing).

2004, $n = 10,000,000,000,000$. Gourdon and Demichel used the Odlyzko-Schönhage algorithm. They also checked a few zeros of much larger height.

These calculations yield strong support in favor of the correctedness of the Riemann hypothesis. However, numerical evidence can be misleading when dealing with arithmetic. For example, the inequality $\pi(x) < \mathrm{li}(x)$ was noted by Riemann for x up to 3×10^6 and now has been verified up to 10^{14}, while it is known (Littlewood) that $\pi(x) > \mathrm{li}(x)$ for a set of values of x of positive logarithmic density and also there is a very large positive constant A such that $\pi(x) - \mathrm{li}(x)$ has a sign change in the interval $[N, AN]$ for every positive integer N. Thus some mathematicians remain skeptical about the value of numerical evidence with respect to the distribution of prime numbers and of the Riemann hypothesis. For example, there is serious evidence that the first sign change of $\pi(N) - \mathrm{li}(N)$ occurs around $N \asymp 1.397 \times 10^{316}$, an astonishingly large number indeed. Thus the true behavior of prime numbers can be observed only in ranges far beyond the computable. However, this phenomenon is now well explained and is in complete accord with the Riemann hypothesis.

Claims for proofs of the Riemann hypothesis abound but so far all that could be checked have proved to be incorrect. Stieltjes announced in 1885 a proof of the Riemann hypothesis based on the estimate $|M(x)| < A\sqrt{x}$ where $M(x) = \sum_{n \le x} \mu(n)$ and said that his proof was "very fatiguing" and difficult and only after a long series of auxiliary results. He also said in 1887 that he had managed to simplify somewhat his proof, but nothing was found in his notes after his death. Jensen in 1899 announced to have proved the Riemann hypothesis using his formula (the Jensen

formula) evaluating the integral of the logarithm of a holomorphic function on a circle in terms of the zeros of the function inside the circle, but again this claim had no follow-up. For all this, and other attempts, the reader may consult with profit the book by Narkiewicz [43].

7. More general zeta and L-functions

Here and in the next sections we define more general zeta and L-functions.

Let K be a number field. The Dedekind zeta function is defined for $\Re(s) > 1$ by

$$\zeta(s, K) = \sum_{\mathfrak{n}} N(\mathfrak{n})^{-s}$$

where the sum range over all integral ideals $\mathfrak{n}$ of O_K and the norm $N(\mathfrak{n})$ is the absolute norm $|O_K/\mathfrak{n}|$. By the uniqueness of the decomposition of an integral ideal into a product of prime ideals and the multiplicativity of the norm, we have an Euler product

$$\zeta(s, K) = \prod_{\mathfrak{p}} \Big(1 - \frac{1}{N(\mathfrak{p})^s}\Big)^{-1}.$$

The analytic continuation of the Dedekind zeta function is not a trivial fact. It was obtained by Hecke, who proved that $\zeta(s, K)$ is meromorphic in the complex plane, with only a simple pole at $s = 1$.

Let $d = [K : \mathbb{Q}]$ and let r_1 be the number of inequivalent real embeddings $K \to \mathbb{R}$ and let r_2 be the number of inequivalent complex embeddings, hence $d = r_1 + 2r_2$, and Δ_K the discriminant of K.

Theorem (Hecke, 1917).

- *The Dedekind zeta function is meromorphic of order* 1 *with a simple pole at* $s = 1$ *with residue* $2^{r_1}(2\pi)^{r_2}Rh/(w\sqrt{|\Delta(K)|})$, *where* r_1 *is the number of real embeddings of* K, r_2 *is the number of pairs of complex embeddings of* K, w *is the order of the group of roots of unity in* K, R *is the regulator of* K, h *is the class number, and* $\Delta(K)$ *the discriminant of* K.
- *The function* $\zeta(s, K)$ *satisfies the functional equation*

$$|\Delta_K|^{\frac{s}{2}}\Big(\pi^{-\frac{s}{2}}\Gamma\Big(\frac{s}{2}\Big)\Big)^{r_1}\Big((2\pi)^{-s}\Gamma(s)\Big)^{r_2}\zeta(s, K) \quad \textit{is invariant by } s \mapsto 1 - s.$$

Hecke's proof is a very clever extension of Riemann's second proof of the functional equation of $\zeta(s)$. Its main ingredients are:

- O_K is a lattice in $\mathbb{R}^{r_1} \times \mathbb{C}^{r_2}$.
- Express $\zeta(s, K)$ as Mellin transform over $\mathbb{R}^{r_1} \times \mathbb{C}^{r_2}$ of a multidimensional theta series.
- Use Poisson summation to obtain the functional equation of the theta series and deduce from it the analytic continuation and the functional equation of $\zeta(s, K)$.

Another proof of the functional equation of the Dedekind zeta function, more in line with Riemann's first proof of the functional equation for $\zeta(s)$, was obtained by Siegel in [50]. He gives the proof only for the case of a totally real field, but mentions that it can be extended to cover the general case (also with Hecke Grössencharacters).

Hecke introduced a new type of character, called *Grössencharacter*, and showed that the corresponding L-functions had similar properties as the Dirichlet L-functions. A Grössencharacter is defined as follows.

- **Absolute values:** For each place v, let $\|x\|_v$ be the associated absolute value: $N(\mathfrak{p})^{-\mathrm{ord}_{\mathfrak{p}}(x)}$ if $v = \mathfrak{p}$, $|x|$ if $v = \mathbb{R}$, $|x|^2$ if $v = \mathbb{C}$.
- **Idèles:** $J = \prod_v K_v^\times$, with the product restricted to elements x with almost every factor $\|x_v\|_v = 1$, together with a suitable topology. Note that $K^\times \subset J$ *via* the diagonal embedding.
- **Hecke Grössencharacter:** A continuous homomorphism $\psi : K^\times \backslash J \to \mathbb{T}$ with image dense in $\mathbb{T}$ (i.e. "Grössen").
- **The conductor:** A place v is *unramified* for ψ if

$$\psi_v(x_v) := \psi((1, \ldots, 1, x_v, 1, \ldots)) = 1 \quad \text{whenever} \quad \|x_v\| = 1.$$

  The conductor $\mathfrak{f}$ of ψ is the ideal

$$\mathfrak{f} = \prod_{\mathfrak{p} \text{ ramified}} \mathfrak{p}^{m_v}$$

  where m_v is the smallest exponent for which $\psi_v(x_v) = 1$ for $x_v \in 1 + \mathfrak{p}^{m_v}$.
- **Grössencharacter of an ideal:** It suffices to define it on prime ideals, as $\psi(\mathfrak{p}) = \psi(\varpi_v)$ with ϖ_v a uniformizer of $\mathfrak{p}$ if $\mathfrak{p} \nmid \mathfrak{f}$, and 0 otherwise.

  For $K_v = \mathbb{R}$ or $\mathbb{C}$, we have

$$\psi_v(x_v) = \Big(\frac{x_v}{|x_v|}\Big)^{m_v} |x_v|^{i\tau_v}$$

where $m_v = 0$ or 1 if $K_v = \mathbb{R}$ and $m_v \in \mathbb{Z}$ if $K_v = \mathbb{C}$.

Hecke proved that a Grössencharacter can be used to define a new type of L-function enjoying properties similar to those for Dirichlet's L-functions.

Theorem (Hecke, 1920).

- *The L-function*

$$L(s, \psi) = \sum_{\mathfrak{a}} \frac{\psi(\mathfrak{a})}{N(\mathfrak{a})^s} = \prod_{\mathfrak{p} \text{ unramified}} \Big(1 - \frac{\psi(\mathfrak{p})}{N(\mathfrak{p})^s}\Big)^{-1}$$

  attached to a non-trivial Grössencharacter ψ is entire of order 1.
- *Let $\Delta_\psi = |\Delta(K)|\, N(\mathfrak{f}_\psi)$ and define*

$$\Lambda(s, \psi) = (\Delta_\psi)^{s/2} \prod_{K_v = \mathbb{R}} \pi^{-\frac{s}{2}} \Gamma\Big(\frac{s}{2} + c_v\Big) \prod_{K_v = \mathbb{C}} (2\pi)^{-s} \Gamma(s + c_v)\, L(s, \psi)$$

where $c_v = (i\tau_v + |m_v|)/2$. *Then*

$$\Lambda(s,\psi) = w(\psi)\Lambda(1-s,\overline{\psi})$$

for some complex number $w(\psi)$ *with* $|w(\psi)| = 1$.

8. Artin L-functions

L/K is a Galois extension of degree d of K with Galois group G. Let $\pi : G \to GL(n,\mathbb{C})$ be a representation of G. The lift $\mathfrak{p}\,\mathcal{O}_L$ to L of a prime ideal $\mathfrak{p}$ of K factors as

$$\mathfrak{p}\,\mathcal{O}_L = \prod_{i=1}^{r} \mathfrak{P}_i^e.$$

Here e is the ramification index, $|\mathcal{O}_L/\mathfrak{P}_i| = |\mathcal{O}_K/\mathfrak{p}|^f$ and $efr = d$.

G acts on $\{\mathfrak{P}_1,\dots,\mathfrak{P}_r\}$ by permutations; the subgroup $G_{\mathfrak{P}}$ fixing $\mathfrak{P}$ is the *decomposition group* of $\mathfrak{P}$. The elements $\sigma \in G_{\mathfrak{P}}$ with

$$\sigma x \equiv x^{N(\mathfrak{p})} \pmod{\mathfrak{P}}$$

form a right- and left-coset $(\mathfrak{P}, L/K)$ (the *Frobenius substitution*) of the *inertia group* $I_{\mathfrak{P}}$ fixing $K(\mathfrak{P}) = O_L/\mathfrak{P}$. Changing $\mathfrak{P}$ into $\eta\mathfrak{P}$ yields $(\eta\mathfrak{P}, L/K) = \eta(\mathfrak{P}, L/K)\eta^{-1}$.

Let $\chi = \mathrm{Trace}(\pi)$ be the character of π and define for $\sigma \in (\mathfrak{P}, L/K)$:

$$\chi(\mathfrak{p}^m) = |I_{\mathfrak{p}}|^{-1} \sum_{\tau \in I_{\mathfrak{p}}} \chi(\sigma^m \tau).$$

Then $\chi(\mathfrak{p}^m)$ is independent of the choices of $\mathfrak{P}$ and σ.

Definition. *The local Artin L-function is* $L_{\mathfrak{p}}(s,\chi,L/K)$ *is*

$$L_{\mathfrak{p}}(s,\chi,L/K) := \exp\left(\sum_{m=1}^{\infty} \frac{1}{m}\chi(\mathfrak{p}^m)N(\mathfrak{p})^{-ms}\right).$$

The global Artin L-function is

$$L(s,\chi,L/K) = \prod_{\mathfrak{p}} L_{\mathfrak{p}}(s,\chi,L/K).$$

Artin L-functions satisfy relations inherited from operations on representations of groups, the Artin formalism. We have

- $L(s,1,L/K) = \zeta(s,K)$.
- **Direct sum:** $L(s,\chi_1+\chi_2,L/K) = L(s,\chi_1,L/K)\,L(s,\chi_2,L/K)$.
- **Restriction:** Let $L' \supset L \supset K$ be Galois extensions. Then

$$L(s,\chi,L'/K) = L(s,\chi,L/K).$$

- **Induction:** Let $L' \supset L \supset K$ be Galois extensions and let χ^* be the character of $G = \mathrm{Gal}(L'/K)$ induced by a character χ of $H = \mathrm{Gal}(L'/L)$. Then
$$L(s, \chi, L'/L) = L(s, \chi^*, L'/K).$$
The character χ^* is the unique character such that $(\chi^*, \psi)_G = (\chi, \psi|_H)_H$ where $(\ ,\ )_G$ is the scalar product on central functions on G normalized with $(1,1)_G = 1$.

Theorem (Artin, 1923). *We have*
$$L(s, \chi, L/K) = \prod_{\mathfrak{p}} \det\left[I - \frac{\pi(\mathfrak{p})}{N(\mathfrak{p})^s}\right]^{-1}$$
where
$$\pi(\mathfrak{p}) = \pi(\sigma)|I_{\mathfrak{p}}|^{-1} \sum_{\tau \in I_{\mathfrak{p}}} \pi(\tau).$$

For the proof, note that $|I_{\mathfrak{p}}|^{-1} \sum_{\tau \in I_{\mathfrak{p}}} \pi(\tau)$ is an idempotent.

An example. Let $G = \{\pm 1\}$, $L/K = \mathbb{Q}(\sqrt{D})/\mathbb{Q}$, D square free. $\Delta = D$ if $D \equiv 1 \pmod 4$, otherwise set $\Delta = 4D$.

Case 1: $p \nmid \Delta$, $(p) = \mathfrak{p}\overline{\mathfrak{p}}$. Here $G_{\mathfrak{p}} = \{1\}$ and $(\mathfrak{p}, G) = (\overline{\mathfrak{p}}, G) = 1$ by Fermat's Little Theorem.

Case 2: $p \nmid \Delta$, (p) prime. Here $G_{\mathfrak{p}} = \{\pm 1\}$, $I_{\mathfrak{p}} = \{1\}$ and $((p), G) = -1$, because if (p) does not split then D is not a quadratic residue $\pmod p$.

Case 3: $p \nmid \Delta$, $(p) = \mathfrak{p}^2$. Here $G_{\mathfrak{p}} = \{\pm 1\}$, $I_{\mathfrak{p}} = \{\pm 1\}$ and $(\mathfrak{p}, G) = \{\pm 1\}$.

Take π to be the regular representation
$$\pi\{1, -1\} = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \right\}.$$
The local factors are:

- $\det \begin{bmatrix} 1 - p^{-s} & 0 \\ 0 & 1 - p^{-s} \end{bmatrix} = (1 - p^{-s})^2$ if p splits;
- $\det \begin{bmatrix} 1 & -p^{-s} \\ -p^{-s} & 1 \end{bmatrix} = 1 - p^{-2s}$ if p remains prime.
- $\det \begin{bmatrix} 1 - \frac{1}{2}p^{-s} & -\frac{1}{2}p^{-s} \\ -\frac{1}{2}p^{-s} & 1 - \frac{1}{2}p^{-s} \end{bmatrix} = 1 - p^{-s}$ if p ramifies.

Then $L(s, \pi, L/K) = \zeta(s)L(s, (\frac{\Delta}{\cdot}))$.

Identifying $L(s, \pi, L/K) = \zeta(s)L(s, (\frac{\Delta}{\cdot}))$ (where $(\frac{\Delta}{\cdot})$ is the Kronecker symbol) with $\zeta(s, L/K) = \zeta(s)L(s, \chi_\Delta)$, where now χ_Δ is the Dirichlet character, is the same as the *quadratic reciprocity law.*

In the general case of abelian extensions L/K, the corresponding result is Artin's reciprocity law of class field theory.

For example, it implies as a special case the famous Kronecker–Weber theorem that every abelian extension of $\mathbb{Q}$ is a subfield of a cyclotomic field.

Artin's Conjecture. *The L-function $L(s, \chi_\pi, L/K)$ associated to a non-trivial irreducible representation π of $\mathrm{Gal}(L/K)$ is an entire function of s of order 1.*

Known only in special cases. For G abelian, it is due to Artin. It is also known for characters expressible as linear combinations with positive coefficients of characters induced by cyclic subgroups of G. Other cases are $\dim(\pi) = 2$ and $\mathrm{Image}(G)$ in $PSL_2(\mathbb{C})$ tetrahedral (Hecke) or octahedral (Langlands, Tunnell). This has been solved only recently (with some mild side conditions) if the image is the simple alternating group A_5, by Taylor, Buzzard, ... (2002).

A closely related simpler problem is

Landau's conjecture. *The zeta function of a number field is divisible by the Riemann zeta function, in the sense that the quotient is entire.*

Even if we do not know the validity of the Artin conjecture in general, we do know that Artin L-functions are meromorphic of order 1, which is a celebrated theorem of R. Brauer and a consequence of a theorem on representations of finite groups.

Theorem (Brauer, 1948). *$L(s, \chi_\pi, L/K)$ is meromorphic of order 1 and satisfies a functional equation (here $\Lambda = L \times \{\Gamma\text{-}factors\}$)*

$$\Lambda(s, \chi_\pi, L/K) = w(\chi_\pi, L/K)\Lambda(1 - s, \widetilde{\chi}_\pi, L/K)$$

where $\widetilde{\chi}_\pi$ is the character of the contragredient of π.

9. Some modular forms

Let $f(z)$ be a holomorphic modular form of weight k for a subgroup $\Gamma < \Gamma(1)$ generated by $z \mapsto z + h$ and $z \mapsto -1/z$:

$$f(\gamma z) = \varepsilon(\gamma)(cz + d)^k f(z), \qquad \gamma = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

with $\varepsilon(\gamma)$ an appropriate set of multipliers (*Nebentypus*). Then $f(z)$ is periodic and has a Fourier expansion

$$f(z) = \sum_{n=0}^{\infty} a_n e^{2\pi i z/h}.$$

By Mellin transforms, $L(s, f) = \sum_{n=1}^{\infty} a_n n^{-s}$. If $a_0 \neq 0$ then it is meromorphic of order 1 with a simple pole at $s = k$ and

$$(2\pi/h)^{-s}\Gamma(s)L(s, f) = w\,(2\pi/h)^{s-k}\Gamma(k - s)L(k - s, f)$$

for $w = i^k\varepsilon([z \mapsto -1/z])$.

Note that the subgroup Γ has only one equivalence class of cusps, namely the cusp at $i\infty$.

There is a remarkable converse theorem to the above statement.

Theorem (Hecke, 1936). *Conversely, given $L(s,f)$ with the above properties one recovers a modular form f of weight k for Γ by setting $a_0 = i^{-k}w$ in case $L(s,f)$ has a pole at $s=k$.*

Example. Take $h=2$,

$$\theta(z) = \frac{1}{2}\sum_{n=-\infty}^{\infty} e^{\pi i n^2 z}.$$

Then $\theta(z)$ is a modular form of weight $\frac{1}{2}$, multiplier 1 and $L(s,\theta) = \zeta(2s)$. The space of such forms for Γ has dimension 1, hence $\zeta(2s)$ is the unique (up to a scalar) Dirichlet series satisfying the same functional equation as $\zeta(2s)$.

Hecke's converse result uses only the cusp at $i\infty$ and his proof fails to extend to the case of a group Γ with many cusps. Weil's new idea when one has more than one cusp: Control of the Fourier expansions at every cusp can be obtained by *twisting* the Dirichlet series with Dirichlet characters.

Theorem (Weil, 1967). *Let $L(s) = \sum a_n n^{-s}$ and write, for a Dirichlet character $\chi \pmod r$:*

$$\Lambda(s,\chi) = (2\pi)^{-s}\Gamma(s)\sum_{n=1}^{\infty} a_n \chi(n) n^{-s}.$$

Suppose there are N, k such that for every primitive $\chi \pmod r$ with $(r,N)=1$ we have that $\Lambda(s,\chi)$ is entire of order 1 and

$$\Lambda(s,\chi) = w_\chi r^{-1}(r^2N)^{\frac{k}{2}-s}\Lambda(k-s,\overline{\chi})$$

with $w_\chi = i^k\chi(N)\tau(\chi)^2$ and $\tau(\chi) = \sum_{n \pmod r} \chi(n)e^{2\pi i n/r}$. Then we have $L(s) = L(s,f)$ with f a holomorphic cusp form for $\Gamma_0(N)$.

Here $\Gamma_0(N)$ is the Hecke congruence subgroup of level N, namely

$$\Gamma_0(N) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z}),\ \ c \equiv 0 \pmod N \right\}.$$

A very important point in Weil's result is the identification of the conductor N of the L-function with the level of the associated Hecke congruence group.

10. Zeta functions over finite fields

Let V be an algebraic variety defined over a finite field $\mathbb{F}_q$. For every algebraic point P of V let $\mathfrak{p}$ be the orbit of P under the action of the Frobenius automorphism $x \mapsto x^q$ and let $\deg(\mathfrak{p})$ be the cardinality of the set $\{\mathfrak{p}\}$.

The Zeta function $Z(t,V)$ is defined by

$$Z(t,V) = \prod_{\mathfrak{p}} \left(1 - t^{\deg(\mathfrak{p})}\right)^{-1}.$$

If $W \to V$ is an unramified finite covering with group G, defined over $\mathbb{F}_q$, then one can define a Frobenius substitution and, correspondingly, an Artin L-function

$L(t, W/V, \chi)$ attached to the covering $W \to V$ and a character χ of a complex representation of G.

A celebrated theorem of Deligne proves a Riemann hypothesis, namely $Z(t, V)$ is a rational function with zeros and poles of absolute values of the type $q^{-m/2}$ with m a positive integer. In his paper, Deligne also proves that the reciprocal of zeros and poles are eigenvalues of the Frobenius automorphism on the l-adic cohomology of V with compact support and l-adic coefficients, thus obtaining exact bounds on the exponents m. Such bounds are of the utmost importance for applications.

The following historical note clarifies why the function $Z(t, V)$ is indeed a 'zeta function' and how it came to be defined in this way. The origin of the function is in the work of Emil Artin and F.K Schmidt showing how these functions can arise in a general algebraic setting not limited to just number fields.

Let $C/\mathbb{F}_q$ be an affine curve with associated function field K and ring R of functions of K regular on C, i.e. only with poles at ∞. Let $\mathfrak{a}$ be an ideal in R. Then $\mathfrak{a}$ factors uniquely into primary ideals $\mathfrak{a} = \prod \mathfrak{p}^{a(\mathfrak{p})}$. If we assign to the prime ideal $\mathfrak{p}$ its norm, namely $N(\mathfrak{p}) = |R/\mathfrak{p}| = q^{\deg(\mathfrak{p})}$, a natural definition for the zeta function, in analogy with the Euler product for the Dedekind zeta function of a number field, is

$$\zeta(s, K) = \prod_{\mathfrak{p}} (1 - N(\mathfrak{p})^{-s})^{-1} = \sum_{\mathfrak{a}} \frac{1}{N(\mathfrak{a})^s}$$

However, this is a function of $t = q^{-s}$ and after this change of variable it becomes $Z(t, C)$. By unique factorization of ideals, it follows that

$$Z(t, C) = \sum_{\mathfrak{a}} t^{\deg(\mathfrak{a})}.$$

The number of ideals with a given degree is computed with the Riemann–Roch theorem; the Riemann part shows that $Z(t, C)$ is a rational function of t and the Roch part yields a functional equation.

Nowadays, the old definition in terms of the variable s has become obsolete.

11. The Hasse–Weil zeta function

The Hasse-Weil zeta function. *Let V/K be a variety over a number field. Then for all except finitely many prime ideals $\mathfrak{p}$ the reduction $\overline{V}_{\mathfrak{p}}$ is defined over the finite field $\mathcal{O}_K/\mathfrak{p}$ and yields a Zeta function $Z(T, \overline{V}_{\mathfrak{p}})$. The global zeta function of V is now*

$$\prod_{\mathfrak{p}} Z(N(\mathfrak{p})^{-s}, \overline{V}_{\mathfrak{p}})$$

where the product runs over all prime ideals where the reduction is defined.

If V'/K is another model of V the associated zeta functions are the same up to finitely many factors.

A serious difficulty is to define 'good factors' even at places of bad reduction and 'good models' for which the zeta function behaves nicely.

Example 1. *Let V be the projective space $\mathbb{P}^n/\mathcal{O}_K$. Then*

$$\zeta(s,V) = \zeta_K(s)\zeta_K(s-1)\cdots\zeta_K(s-n).$$

Example 2 (Deuring, 1953-57). *Let E be an elliptic curve over a number field L, with complex multiplication in the imaginary quadratic field $K = End(E) \otimes \mathbb{Q}$. Assume $K \subset L$. Then there is a model of E/L and a Hecke Grössencharacter ψ of L such that*

$$\zeta(s,E) = \zeta(s,L)\zeta(s-1,L)L\Big(s-\frac{1}{2},\psi\Big)^{-1}L\Big(s-\frac{1}{2},\overline{\psi}\Big)^{-1}.$$

Example 3 (Taniyama, 1957). *Example* 2 *extends to abelian varieties of CM type.*

The theory of these more general zeta and L-functions received a new impetus from the Shimura–Taniyama conjecture. A precursor of the conjecture can be traced to a paper of Eichler (1953), where he proved the functional equation for zeta functions of modular curves $X_0(N) = \mathcal{H}/\Gamma_0(N)$ of genus 1 (there are only finitely many of them), by showing that in this case $L(s, X_0(N))$ is the Mellin transform of a cusp form of weight 2. Shimura extended this to certain other cases, namely from elliptic factors defined over $\mathbb{Q}$ of the Jacobian of the modular curve $X_0(N)$. On the basis of this evidence, Taniyama made a bold suggestion that this held in general.

Conjecture (Taniyama (1955), Shimura, Weil, ...). *Every elliptic curve over $\mathbb{Q}$ is uniformized by a cusp form of weight* 2 *for $\Gamma_0(N)$. Equivalently, every elliptic curve $E/\mathbb{Q}$ with conductor N admits the modular curve $X_0(N)$ as a ramified covering.*

Weil's contribution to the conjecture was, as previously mentioned, the identification of the level of the modular curve with the conductor of the elliptic curve. This made the conjecture verifiable, at least in principle, for every fixed N.

This was proved at last by Wiles and Taylor (1995) in the semistable case (namely multiplicative bad reduction only) and in general by Breuil & Conrad & Diamond & Taylor (2001). A consequence of the theorem of Wiles and Taylor is a proof of Fermat's Last Theorem, after more than 300 years and thousands of failed attempts by amateurs and mathematicians alike.

12. The Rosetta Stone of L-functions

The famous Rosetta Stone carries an identical text with parallel inscriptions in hieroglyphs, demotic and Greek. It is associated with the famous Egyptologist Champollion (1790-1832), who used it, after he had deciphered demotic, as the starting point for reading hieroglyphs.

He recognized on the Rosetta Stone the name Ptolmys in Greek and demotic, and from there he identified the same name in hieroglyphs, written in a cartouche. Three years later, in 1821, while studying corresponding hieroglyphic and Greek texts on an obelisk transported to England by Giovanni Belzoni (1778-1823), he recognized the name Kliopadra, thus getting the values of twelve hieroglyphs. From there, he was able to complete the monumental task he had started in 1808 at the age of eighteen.

Zeta and L-functions, starting from the mysterious first object found by Riemann, have by now grown into an even more mysterious set of objects that encode the most delicate aspects and questions of arithmetic.

André Weil, who greatly contributed to the subject, loved to compare our understanding of these functions to the task of deciphering hieroglyphics through the help of the Rosetta Stone. He was fond of pointing out that zeta and L-functions appear in three different flavors, as the parallel texts in three different languages inscribed in the Rosetta Stone, although as yet no Champollion of mathematics has appeared on the stage.

The three different ways of obtaining L-functions may be summarized as follows.

- **Motivic writing:** L-functions can be defined from geometry.
- **Galois writing:** L-functions can be defined from finite dimensional representations of the absolute Galois group $\mathrm{Gal}(\overline{K}/K)$ acting on a vector space V.
- **Automorphic writing:** L-functions can be defined globally from automorphic forms on algebraic groups modulo discrete subgroups.

After the pioneering work of Hasse and Weil, fundamental contributions to the geometric construction of L-functions came from the work of Grothendieck and his school, in particular with the systematic introduction of deep cohomological tools such as l-adic cohomology, various kinds of p-adic cohomology theories, the Lefschetz Trace formula, and Grothendieck's theory of motives.

The Galois point of view originates with Artin's construction of L-functions.

The automorphic point of view consists in obtaining L-functions as Mellin transforms of general automorphic functions.

Artin's method using representation theory to construct L-functions and the automorphic method received new impetus from Langlands's work around 1970. There, Langlands vastly extended the relation between automorphic forms and L-functions and set the foundation for a general theory. This can be seen as a deep fusion and extension of Artin's as well as Hecke's ideas. A Hecke character is nothing else than a representation of $GL(1,\mathbb{A})$ in the space of continuous functions on $GL(1,K)\backslash GL(1,\mathbb{A})$. An automorphic representation of the general linear group is an irreducible component π of a representation of $GL(n,\mathbb{A})$ on the space of continuous functions on the homogeneous space $GL(n,K)\backslash GL(n,\mathbb{A})$ (with technical conditions). One can then attach to π an L-function, with functional equation $L(s,\pi) = w(\pi)L(1-s,\widetilde{\pi})$ with $\widetilde{\pi}$ the contragredient. Also π decomposes as a tensor product $\otimes\pi_v$ of local components, yielding an Euler product for $L(s,\pi)$ with standard factors of degree n.

13. Deciphering the Rosetta Stone

The deciphering of the Rosetta Stone is still incomplete, but large pieces are emerging on a continuous basis, showing that all three ways of generating L-functions are closely related and probably lead to a single class of objects.

- **Automorphic $\supset$ Galois:** This is a key step with very deep implications. For $GL(1)$, it is founded in Artin's abelian reciprocity law of class field theory, a vast generalization of the quadratic reciprocity law. For $GL(2)$ it is known for dihedral (reduces to $GL(1)$), tetrahedral, and octahedral representations and many (but not yet all) icosahedral representations, *i.e.* Image$(G) = S_3, S_4, A_5$ (Artin, Langlands, Langlands and Tunnell, Taylor, Buzzard, Skinner, ...).
- **Galois = Motivic:** This is also a key step, understood in very few cases. It is known for $GL(1)$ (Artin, Hecke) and a few other cases.
- **Motivic $\subset$ Automorphic:** Also known in very few cases. Surely there are more automorphic L-functions than motivic L-functions coming from geometry, because of L-functions associated to eigenfunctions of the hyperbolic Laplacian on $\Gamma\backslash\mathbb{H}$ which are simultaneously Hecke eigenfunctions at all primes (the Maass wave forms).

  It was proved by Wiles and Taylor for semistable elliptic curves $E/\mathbb{Q}$ and then for all elliptic curves $E/\mathbb{Q}$ (the Taniyama–Shimura conjecture).

14. Applications and families of L-functions

If

$$L(s,f) = \prod_v \big(1-\omega_1(\mathfrak{p})N(\mathfrak{p})^{-s}\big)^{-1}\big(1-\omega_2(\mathfrak{p})N(\mathfrak{p})^{-s}\big)^{-1}$$

the symmetric k-th power is given by

$$L(s, Sym^k f) = \prod_v \prod_{j=0}^{k} \big(1-\omega_1(\mathfrak{p})^j\omega_2^{k-j}(\mathfrak{p})N(\mathfrak{p})^{-s}\big)^{-1}.$$

A deep conjecture is that these symmetric powers have analytic continuation and a functional equation. A first success of the theory was to prove this conjecture for $k = 2, 3, 4$ ($k = 2$ by Rankin and Selberg for classical cusp form and Gelbart and Jacquet (1979), for $k = 3$ and 4 by Kim and Shahidi (2000)).

The work of Wiles and Taylor provided new ideas that were applied with success by Taylor (following joint work with L. Clozel, M. Harris, and N. Shepherd-Barron) in the case of symmetric powers arising from a non-CM elliptic curve over $\mathbb{Q}$, semistable at at least one place. They were able to prove the expected functional equation and the analytic continuation, but only as a meromorphic function. This is reminiscent of the similar problem encountered with the Artin conjecture. Indeed, this is much more than a superficial similarity, because a key final argument in the proof has pretty much the same structure as what occurred with Brauer's proof that Artin L-functions were meromorphic. The problem of showing that these functions are entire remains wide open.

The removal of the extra condition and the extension of this result to GL_2 L-functions has now been achieved by Taylor and his collaborators (2009).

A consequence of this work is the unconditional proof of a conjecture of Sato and Tate giving the distribution of the traces $a_p = \text{Trace}(\phi : E_p)$ of the Frobenius

automorphism ϕ on the reduction E_p of E modulo p of a non-CM elliptic curve $E/\mathbb{Q}$. It follows Wigner's Semicircle Law, namely for $-1 < a < b < 1$ we have

$$\#(p \le x \,:\, 2\sqrt{p}\,a < a_p < 2\sqrt{p}\,b) \sim \Big(\frac{2}{\pi}\int_a^b \sqrt{1-u^2}\,\mathrm{d}u\Big)\frac{x}{\log x}.$$

The distribution of traces in the CM case is different.

Now after the pioneering work of Katz and Sarnak we understand that zeta and L-functions can be grouped into distinct families according to their fine analytic behavior. Extensive computer explorations of the behavior of these functions show that these functions present a high degree of randomness, but also that this behavior does not always follow the standard Gaussian law. The following examples describe the kind of statistical analysis that one may do when studying a discrete sequence of values and the associated correlations.

Let $\{\lambda_j\}$ be a real sequence, $\lambda_j \sim j$. Consider the gaps $\Delta_j = \lambda_{j+1} - \lambda_j$ and set

$$\mu_1(N)[a,b] = \frac{1}{N}\#\{j \,:\, \Delta_j \in [a,b]\}, \quad 0 \le j < N.$$

Quite often one finds a Cauchy-Poisson distribution $\mu_1(N) \to e^{-x}\mathrm{d}x$.

Example: This is expected for the sequence $\{p_j/\log p_j\}$, although this remains open.

Example: Again, expected for the sequence $\{\pi 2^{-9/4}(m^2+\sqrt{2}n^2)\}$ and, again, completely open.

So the question is to find the type of randomness exhibited by zeta and L-functions. The simplest instance is

Problem: What is the behavior for $\{\frac{1}{2\pi}\gamma\log\gamma\}$, where $\zeta(\frac{1}{2}+i\gamma)=0$ and $\gamma>0$?

In 1973, Hugh Montgomery, while at the Institute for Advanced Study, began the study of the first order spacings $\{(\gamma_{n+1}-\gamma_n)\log|\gamma_n|\}$, obtaining some partial intriguing results. During a discussion with Freeman Dyson over tea-time, Dyson recognized that Montgomery's results matched the distribution of eigenvalues of random unitary matrices, the so-called Gauss Unitary Ensemble or, briefly, GUE. Thus the connection of random matrix theory with the Riemann zeta function was born. Extensive calculation by Odlyzko with zeros of $\zeta(s)$ around 10^{20} showed a total agreement with the prediction of Montgomery and Dyson. The only problem is: Why is this so?

The eigenvalue distribution of random unitary matrices is as follows. For $A \in U(N)$, consider the eigenvalues $z_1, \dots, z_N$ of A on the unit circle, ordered by increasing argument $\pmod{2\pi}$. Let

$$\mu_{k,N}(A)[a,b] = \frac{1}{N}\#\Big\{j : \frac{N}{2\pi}\arg(z_{j+k}/z_j) \in [a,b]\Big\}.$$

Then the following holds.

Theorem (Gaudin, 1961, Katz and Sarnak, 1999). *We have*

$$\int_{U(N)} \mu_{k,N}(A)\,\mathrm{d}A \to \mu_k([a,b])$$

(here $\mathrm{d}A$ *is the normalized Haar measure) with*

$$\mathrm{d}\mu_k = \frac{\mathrm{d}^2}{\mathrm{d}s^2}\Big(\sum_{j=0}^{k-1}\frac{k-j}{j!}\Big(\frac{\partial}{\partial T}\Big)^j \det\big(I+TK(s)\big)\Big|_{T=1}\Big)\,\mathrm{d}s$$

and $K(s)$ *the operator defined by*

$$K(s)\phi(x) = \int_{-\frac{s}{2}}^{\frac{s}{2}} \frac{\sin(x-y)}{\pi(x-y)}\phi(y)\,\mathrm{d}y.$$

In a major work, Katz and Sarnak [37] analyzed the distribution of arguments of the characteristic roots of $Z(t, C/\mathbb{F}_q)$, where C runs over an algebraic family of curves over $\mathbb{F}_q$, as $q = p^\nu$ increases with $\nu \to \infty$. Katz and Sarnak have shown that in this function field analogue the prediction is verified for families for which Deligne's theory applies. The associated groups are nothing else than the monodromy groups of the families. However, so far no one has been able to find something that would play the role of the monodromy group for the families associated with classical L-functions, adding another mystery to the theory.

There are also formulas for the classical groups $SU(N), O(N), SO(N), USp(N)$. The amazing thing is that certain families of L functions seem to follow a same correlation, always corresponding to the eigenvalue distribution of families of Lie groups. For example, the distribution of the j-th zero of $L(s,\chi)$, χ a primitive quadratic character, follows the USp prediction. Instead, the j-th zero of $L(s, E\otimes\chi)$, E an elliptic curve, follows the O prediction (two cases, according to the sign $\pm$ in the functional equation, corresponding to the two connected components $O^\pm$ of O). Note however that there are intermediate cases and the prediction of the correct behavior is not quite automatic, see the paper [25] by Duñez and Miller.

Question: For families of classical L-functions, what should take the role of monodromy in order to explain how these laws arise?

Question: Is this phenomenon peculiar to L-functions or is it instead the expression of a 'universality law' which holds in a much wider context?

Question: Is there a way of formulating a program to prove GRH following the blueprint of Deligne in a new context?

In the function field case, Ulmer [54] in 2002 found that when $d = p^n + 1$ the non-isotrivial elliptic curve $y^2 + xy = x^3 - t^d$ over $\mathbb{F}_p(t)$ has rank as big as $(p^n-1)/(2n)$, hence of order at least $\log N/\log\log N$ where N is the conductor of the curve. This result at first seems to confirm, *via* a Birch and Swinnerton-Dyer conjecture, the early prediction *via* random matrix theory that the maximum order of $\log^+|\zeta(\frac{1}{2}+it)|$ is $O(\log|t|/\log\log|t|)$.

On the other hand, more recent predictions, still *via* random matrix theory but with much more refined models, seem to indicate that the maximum order of $\log^+ |\zeta(\frac{1}{2}+it)|$ is $O\big(\sqrt{(\log|t|)\log\log|t|}\big)$.

The zeta function appears to grow very slowly along the critical line, which agrees better with the second prediction. It turns out that the first prediction stays too much to the surface and fails to distinguish between the concepts of analytic conductor and arithmetic conductor. The vertical distribution of zeros of the zeta function (or general L-functions) is associated to the analytic conductor, essentially the height T of the imaginary parts of zeros. Instead, the Birch and Swinnerton-Dyer conjecture is associated with the order of zero of the L-function $L(s, E/\mathbb{Q})$ of an elliptic curve $E/\mathbb{Q}$. The multiplicity of this zero is controlled by the size of the arithmetic conductor N of the curve, determined by the places of bad reduction of a minimal model of E, which is closely related to the discriminant of the elliptic curve (in Weierstrass form, the discriminant of the associated cubic polynomial). On an appropriate Riemann hypothesis, this multiplicity is bounded by $O(\log N/\log\log N)$, which squarely fits with Ulmer's result. The most plausible analogue of Ulmer's result is the prediction of the existence of elliptic curves $E/\mathbb{Q}$ with arbitrarily large conductor N, such that the associated L-function $L(s, E/\mathbb{Q})$ has a zero at $s = 1$ of order $\log N/\log\log N$; on the Birch and Swinnerton-Dyer conjectures, this would be equivalent to the statement that $E/\mathbb{Q}$ has rank of order $\log N/\log\log N$ infinitely often.

While the analogy with the function field case is often a good predictor for the classical case too, it should be taken with much care and not to be considered as a wholesale blueprint for conjectures in the classical case.

The recent proof by Lafforgue (2002) that **motivic** = **automorphic** in the function field case gives support to the hoped for deciphering of the new Rosetta Stone, namely that **motivic** $\subset$ **automorphic** also holds in the classical case.

15. Zeta-like functions without a Riemann hypothesis

The function

$$Z(s) := \frac{1}{2}\sum_{(m,n)\in\mathbb{Z}^2\setminus\{(0,0)\}} \frac{1}{(m^2+5n^2)^s}$$

is the zeta function of the class of principal ideals in $\mathbb{Q}(\sqrt{-5})$. It satisfies the functional equation

$$\Big(\frac{2\pi}{\sqrt{20}}\Big)^{-s}\Gamma(s)Z(s) = \Big(\frac{2\pi}{\sqrt{20}}\Big)^{-(1-s)}\Gamma(1-s)Z(1-s),$$

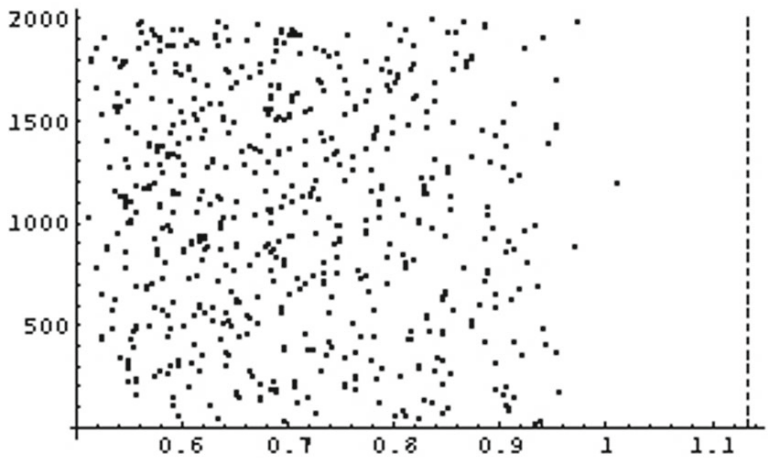
the same as for $\zeta(s, \mathbb{Q}(\sqrt{-5}))$.

This function shares some properties of zeta functions of a quadratic field. Besides the functional equation, we note:

- Distribution of zeros: The number of complex zeros up to height T is
$$\frac{T}{\pi}\log\frac{T}{\pi}-\frac{T}{\pi}+\log\sqrt{5}\,\frac{T}{\pi}+O(\log T).$$
- A positive percentage of the complex zeros of $Z(s)$ lie on the line $\Re(s)=\frac{1}{2}$ (Selberg, unpublished, 1997).
- On GRH and a 'weak' pair correlation conjecture, almost all complex zeros of $Z(s)$ have $\Re(s)=\frac{1}{2}$, see [8].

However, the Riemann hypothesis fails badly:

- $Z(s)$ has infinitely many complex zeros with real part greater than 1.
- The zeros of $Z(s)$ with $\Re(s)>\frac{1}{2}$ admit a distribution function. More precisely, if $\frac{1}{2}<\alpha<\beta$ we have
$$\#\{\rho: Z(\rho)=0,\ \alpha<\Re(\rho)<\beta,\ \ T<\Im(\rho)<2T\}\ \ \sim\Big(\int_\alpha^\beta \mu(\tau)\,\mathrm{d}\tau\Big)T$$
as $T\to\infty$, where $\mu(\tau)\geq 0$ admits right- and left- derivatives at every point (see [5], [39]) and $\mu(\tau)$ is not identically 0 on $1<\tau<\infty$.
- Zeros with $\Re(\rho)>1$ (see [10]):
$$\sup\{\Re(\rho)\ :\ Z(\rho)=0\}=\sigma^*=1.1339063220928286\ldots.$$
- For some constant $C_1>0$ the function $Z(s)$ has infinitely many zeros in the region
$$\sigma>\sigma^*-C_1\,\frac{(\log\log(|t|+3))^{1-2/\sigma^*}}{(\log(|t|+3))^{1-1/\sigma^*}}.$$
- For some constant $C_2>0$, $C_3>0$ the function $Z(s)\neq 0$ in the region
$$\sigma>\sigma^*-\frac{C_2}{(|t|+3)^{C_3}}.$$
- A plot of some zeros with $\Re(\rho)>0.5$:



527 zeros of $Z(s)$ with $\sigma>0.5$ and $0<\gamma<2000$.
$\rho_{303}=1.0110493340435507\ldots+i\,1193.9464045288461075\ldots$

This example is a linear combination of the Dedekind zeta function of the quadratic field $\mathbb{Q}(\sqrt{-20})$ and the Hecke L-function associated to the non-trivial quadratic character of the class-group of the field, which has class number $h(-20) = 2$. All the results obtained extend to the few other imaginary quadratic fields with class number 2, but the determination of zero-free regions to the left of the critical abscissa σ^* remains to be done if the class number is 3 or more.

The example above is associated to zeta and L-functions of $GL(2)$-type. The first example of a function of $GL(1)$-type with a similar behavior was obtained by Davenport and Heilbronn [21] in 1936; it is a well-chosen linear combination of the two Dirichlet L-functions $L(s,\chi)$ and $L(s,\overline{\chi})$ for the complex character

$$\chi(n) = (1, i, -i, -1, 0) \quad (n \mod 5).$$

The function $L(s,\chi)$ satisfies the functional equation

$$\big(\frac{\pi}{5}\big)^{-s/2}\Gamma\Big(\frac{s+1}{2}\Big)L(s,\chi) = \varepsilon_\chi\big(\frac{\pi}{5}\big)^{-(1-s)/2}\Gamma\Big(\frac{(1-s)+1}{2}\Big)L(1-s,\overline{\chi})$$

with $|\varepsilon_\chi| = 1$. If we take

$$f(s) = \big(\varepsilon_\chi^{-\frac{1}{2}}L(s,\chi) + \varepsilon_\chi^{\frac{1}{2}}L(s,\overline{\chi})\big)/(\varepsilon_\chi^{-\frac{1}{2}} + \varepsilon_\chi^{\frac{1}{2}})$$

then the function considered by Davenport and Heilbronn is

$$f(s) = \sum_{n=1}^{\infty} \frac{a_n}{n^s}$$

where $a_n = (1, \tau, -\tau, -1, 0)$ $(n \mod 5)$ and $\tau = \sqrt{1+\phi^2} - \phi$ with ϕ the golden section.[8] We have the functional equation

$$\big(\frac{\pi}{5}\big)^{-s/2}\Gamma\Big(\frac{s+1}{2}\Big)f(s) = \big(\frac{\pi}{5}\big)^{-(1-s)/2}\Gamma\Big(\frac{(1-s)+1}{2}\Big)f(1-s).$$

A full analysis of the distribution of solutions of an equation $Z(s) = a$ for a zeta-like function $Z(s)$ was initiated by Bohr. A complete theory was developed by Bohr and Jessen [5], Borchsenius and Jessen [11], and more recently refined and applied by Yoonbok Lee [39] to specific interesting cases.

16. The Explicit Formula

Riemann's exact formula can be done in a wider setting by attaching smooth weights to primes. More precisely, prime powers $n = p^m$ have a natural arithmetical weight given by the von Mangoldt function $\Lambda(n)$; the 'good' Explicit Formula (now written in capitals) expresses the quantity $\sum \Lambda(n)f(n)$ in terms of the zeros of the Riemann zeta function, for a rather general class of weight functions $f(n)$. This extension of Riemann's formula was worked out in several papers by Guinand in the thirties and the forties (see his 1942 paper [34]), often under the assumption of the Riemann hypothesis in order to derive new applications from his formulas.

[8] Titchmarsh [52] (correctly) gives $(\sqrt{10-2\sqrt{5}}-2)/(\sqrt{5}-1)$ for the same number.

In a well-known paper in 1952, André Weil [58] obtained a sweeping extension of Riemann's formula, valid under very general conditions. The characteristic of Weil's formula was that, under the proper identification of certain key quantities, it applied in a universal way to all zeta and L-functions, including those arising from function fields over a finite field. Moreover, Weil formulated the Riemann hypothesis as the positivity of an associated hermitian form, defined in a natural way.

An interesting point of Weil's formulation of the Riemann hypothesis was that, even if did not shed new light in the classical case, in the geometric case over finite fields the required positivity was an immediate consequence of the so-called algebraic index theorem for divisors on a projective non-singular algebraic surface, which states that the quadratic form determined by the intersection number of numerical equivalence classes of divisors has signature $(+,-,\dots,-)$ with only one positive square. In Weil's case, this was needed only for the surface $C \times C$, the product of the curve with itself.

From the functional equation, it is more symmetrical and natural to work with the function

$$\zeta^*(s) = \pi^{-\frac{s}{2}}\Gamma\Big(\frac{s}{2}\Big)\zeta(s)$$

rather than $\zeta(s)$ alone. It turns out that one can attach to the factor $\pi^{-\frac{s}{2}}\Gamma\big(\frac{s}{2}\big)$ also a contribution, depending on the weight $f(x)$, that leads to a version of the Explicit Formula which admits a remarkable generalization to all zeta and L-functions.

The idea behind the Explicit Formula is very simple and is nothing else than an application of the Mellin transform and of the functional equation.

Consider a meromorphic function of finite order $Z(s) = \sum a_n n^{-s}$ with a functional equation $G(s)Z(s) = G(1-s)Z(1-s)$ where G is a product of exponentials and Gamma function factors. Let $f(x)$ (the weight) be defined on $(0,\infty)$ and be rapidly decreasing at 0 and ∞ with all its derivatives of any order.

Denote also by

$$\widehat{f}(s) = \int_0^\infty f(x)x^{s-1}\mathrm{d}x$$

its Mellin transform and by

$$\widetilde{f}(x) = \frac{1}{x}f\Big(\frac{1}{x}\Big)$$

the involution such that

$$\widehat{\widetilde{f}}(s) = \widehat{f}(1-s).$$

Consider now the inverse Mellin transform

$$\frac{1}{2\pi i}\int_{(c)} \frac{\mathrm{d}}{\mathrm{d}w}\log(G(w)Z(w))\,\widehat{f}(w)x^{-w}\mathrm{d}w$$

where $x > 0$ and c is so large that the formal Dirichlet series $(Z'/Z)(s)$ is uniformly and absolutely convergent for $\Re(s) \geq c$. Then if $(Z'/Z)(s) = \sum b_n n^{-s}$ we have

$$\begin{aligned}\frac{1}{2\pi i}\int_{(c)} \frac{\mathrm{d}}{\mathrm{d}w} \log Z(w)\, \widehat{f}(w) x^{-w} \mathrm{d}w &= \sum_n b_n \frac{1}{2\pi i}\int_{(c)} \widehat{f}(w)(nx)^{-w}\mathrm{d}w \\ &= \sum_n b_n f(nx),\end{aligned}$$

all steps, including term-by-term integration, being easily justified because $\widehat{f}(w)$ is rapidly decreasing at ∞.

The next step consists in moving the line of integration to the left, letting $c \to -\infty$. In doing so, we pick up the residues of the meromorphic function which is the logarithmic derivative of GZ.

This must be done with some care, namely integrating over a rectangle with vertices at $(c - iT_\nu, c + iT_\nu, -c + iT_\nu, -c - iT_\nu)$ along a suitable sequence $T_\nu \to \infty$ avoiding passing too close to a singularity. Polynomial growth of $Z(s)$ is usually required at this step.

The integral over the line $\Re(s) = -c$ now is switched to an integral over $\Re(s) = 1 + c$ by making the change of variable $w \mapsto 1 - w$ and using the functional equation for GZ. This brings in the integral of the logarithmic derivative of $GZ(w)$ times $\widehat{f}(1 - w)$, which has inverse Mellin transform $\widetilde{f}(x)$. By doing this, we obtain

$$\sum b_n\big(f(n) + \widetilde{f}(n)\big) + \ \textit{integrals involving}\ \ G = \sum \textit{residues}.$$

The residues come from a logarithmic derivative, hence they come from simple poles with multiplicity m where m is the order of vanishing of GZ (negative if GZ has a pole, positive if it has a zero).

The passage to the limit $c \to \infty$ requires a bit of attention when dealing with the Gamma factors and it is done by "regularization" to remove infinities from both sides. The conclusion for $\zeta(s)$ is

$$\begin{aligned}&\sum_{n=1}^{\infty} \Lambda(n)\Big\{f(n) + \widetilde{f}(n)\Big\} \\ &+ (\log 4\pi + \gamma) f(1) + \int_1^\infty \Big\{f(x) + \widetilde{f}(x) - \frac{2}{x} f(1)\Big\} \frac{\mathrm{d}x}{x - x^{-1}} \\ &\qquad = \widehat{f}(0) - \sum_\rho \widehat{f}(\rho) + \widehat{f}(1)\end{aligned}$$

valid for a large class of weights $f(x)$, for example of class C^1 except for finitely many discontinuities of the first kind and f and f' rapidly decreasing at 0 and ∞.

The general form of the explicit formula for number fields is as follows.

$$\sum_{\mathfrak{a}} \Lambda(N(\mathfrak{a}))\Big\{f(N(\mathfrak{a})) + \widetilde{f}(N(\mathfrak{a}))\Big\} - r_1 W_{\mathbb{R}}(f) - r_2 W_{\mathbb{C}}(f)$$
$$- (\log|\Delta(K)|)f(1) = \widehat{f}(0) - \sum_{\rho} \widehat{f}(\rho) + \widehat{f}(1)$$

where $W_{\mathbb{R}}$ and $W_{\mathbb{C}}$ are two distributions such that their inverse Mellin transforms are the multiplier $\log|w|$ or $2\log|w|$ plus a continuous part. More precisely, their inverse Mellin transforms are $-\log\pi + \Re\{\Gamma'/\Gamma(\frac{1}{2}w)\}$ and $-2\log(2\pi) + 2\Re\{\Gamma'/\Gamma(w)\}$ respectively.

There are many applications of the Explicit Formula.

The deduction of the Prime Number Theorem in the Čebyshev form

$$\psi(x) = \sum_{n \le x} \Lambda(n) \sim x$$

was done, on the assumption of the Riemann hypothesis, by von Mangoldt in 1894. Helge von Koch (1901) proved that the Riemann hypothesis implies $\psi(x) = x + O(\sqrt{x}(\log x)^2)$. The best numerical implication up to date is by Schoenfeld (1976):

$$RH \Leftrightarrow |\psi(x) - x| < \frac{1}{8\pi}\sqrt{x}(\log x)^2 \quad for \quad x > 73.2.$$

Note that if θ is the lowest upper bound for the real part of the zeros of $\zeta(s)$ then, for ever fixed $\varepsilon > 0$, we have $|\psi(x) - x| > x^{\theta-\varepsilon}$ for a set of values of x of logarithmic density 1, hence if the Riemann hypothesis is false (i.e. $\theta > \frac{1}{2}$) the function $\psi(x) - x$ is 'usually' of much higher order than a random deviation of order $\sqrt{x}$.

Using the Explicit Formula and the counting $N(T) = \frac{T}{2\pi}\log\frac{T}{2\pi} - \frac{T}{2\pi} + O(\log T)$ one can show, on the Riemann hypothesis, that

$$\psi(x) = x - 2\sqrt{x}\sum_{|\gamma|<T} \Re\Big(\frac{x^{i\gamma}}{\frac{1}{2}+i\gamma}\Big) + O\Big(\frac{x}{T}(\log xT)^2\Big) + O(\log x).$$

Now the sum $\sum 1/|\frac{1}{2} + i\gamma|$ is divergent and if for suitable x one could align the arguments of $\frac{x^{i\gamma}}{\frac{1}{2}+i\gamma}$ in a fixed direction then one would be able to detect a large deviation of $\psi(x) - x$ from $\sqrt{x}$. Riemann was probably aware of this, when he suggested "*It would be interesting, in a further development, to study the influence of each periodic term contained in the given expression for the totality of prime numbers.*"

Littlewood (1914) proved that $\pm(\psi(x) - x) > C\sqrt{x}\log\log\log x$ infinitely often. Montgomery conjectured (1980) that the true maximum order of $\psi(x) - x$ is $\sqrt{x}(\log\log\log x)^2$. It is now conjectured that the first change of sign of $\pi(x) - \mathrm{li}(x)$ is around 1.398×10^{316}, well beyond computer range!

One can prove that on the Riemann hypothesis and the further assumption that the numbers γ with $\gamma > 0$ are linearly independent over $\mathbb{Q}$ then the function

$(\psi(e^t)-e^t)/e^{t/2}$ has a limiting distribution with associated probability measure $\mu(t)$, with Fourier transform

$$\widehat{\mu}(\lambda) = \prod_{\gamma>0} J_0\left(\frac{2\lambda}{\sqrt{\frac{1}{4}+|\gamma|^2}}\right)$$

with $J_0(x)$ the Bessel function of the first kind. A thorough study of the corresponding problem for comparing primes in arithmetic progressions was done by Rubinstein and Sarnak (1994). In particular, they gave strong evidence that the density of the set of t for which $\pi\big(e^t\big) < \mathrm{li}\big(e^t\big)$ is about $0.99999973\ldots$, giving a good explanation why $\pi(x) < \mathrm{li}(x)$ in the range of all numerical calculations (which go only up to 10^{23}).

As shown by Weil, an important criterion for the Riemann hypothesis comes from a rewriting of the Explicit Formula. Define a linear functional

$$\begin{aligned} T[f] &:= \int_0^\infty \Big\{f(x)+\widetilde{f}(x)\Big\}\mathrm{d}x - \sum_{n=1}^\infty \Lambda(n)\Big\{f(n)+\widetilde{f}(n)\Big\} - W_{\mathbb{R}}(f) \\ &= \sum_\rho \widehat{f}(\rho) \end{aligned}$$

and let

$$(f*g)(x) = \int_0^\infty f(x)g\Big(\frac{x}{y}\Big)\,\frac{\mathrm{d}y}{y}$$

be the multiplicative convolution. We have *Weil's criterion for RH*:

Theorem (Weil, 1954). *The Riemann hypothesis is equivalent to the statement that the hermitian form $\langle f, f\rangle = T[f*\overline{f}]$ is positive definite on the vector space of complex valued smooth functions with compact support in $(0,\infty)$.*

In fact, Weil proved that this criterion is valid in his more general setting.

It is known that this is so if the support is restricted to a small neighborhood of 1 (Yoshida [55] in 1992, Burnol [13] in 2000, Bombieri [7] in 2001, with different proofs).

Another criterion was found by Li (1997):

$$RH \quad \Leftrightarrow \quad \sum_\rho \Big[1-\Big(1-\frac{1}{\rho}\Big)^n\Big] > 0$$

for all positive integers n. Here ρ runs over all non-trivial zeros of $\zeta(s)$. It has been shown (Lagarias and Bombieri, [9]) that Li's formula can be written as the limit of $\langle f_{n,\varepsilon}, f_{n,\varepsilon}\rangle$ for $\varepsilon\to 0$ for an explicit test function $f_{n,\varepsilon}$, hence Li's criterion shows that it suffices to verify Weil's criterion for just this special class of functions.

Theorem. *In order to verify Weil's criterion it suffices to test it, in the limit for* $\varepsilon \to 0$*, for the functions*

$$f_{n,\varepsilon}(x) = \begin{cases} 0 & \text{if} \quad 0 < x < \varepsilon \\ L_n^{(1)}(-\log x) & \text{if} \quad \varepsilon < x < 1 \\ 0 & \text{if} \quad 1 < x < \infty \end{cases}$$

for $n \in \mathbb{N}$*. Here* $L_n^{(1)}(x)$ *is the generalized Laguerre polynomial.*

17. The Zeta function of a curve

The zeta function of an affine curve over a finite field is a direct analog of the zeta function of a number field, but the better properties (functional equation, etc.) become apparent when one passes to the projective non-singular model which is uniquely defined up to isomorphism, for example using normalization.

Let $k := \mathbb{F}_q$ and let C/k denote such a model of the function field of $k(C)$ (the field of rational function on C defined over k). Then the Zeta function $Z(t, C/k)$ is given by

$$Z(t, C/k) = \prod_{\mathfrak{p}} \left(1 - t^{\deg \mathfrak{p}}\right)^{-1} = \exp\left(\sum_{n=1}^{\infty} |C(k_\nu)| \, \frac{t^n}{n}\right)$$

where $C(k_\nu)$ is the set of points of C defined over $k_\nu = \mathbb{F}_{q^\nu}$, the unique extension of k of degree ν.

The first main result (Artin, F.K. Schmidt, Hasse) is that $Z(t, C/k)$ is a rational function of t and satisfies a functional equation.

Theorem. *Let* C/k *be a projective non-singular curve over* k *and let* g *be its genus. Then we have*

$$Z(t, C/k) = \frac{P(t)}{(1-t)(1-qt)}$$

where $P(t)$ *is a polynomial of degree* $2g$ *of the type*

$$P(t) = \prod_{h=1}^{2g} (1 - \omega_h t)$$

and where $\omega_{h+g} = q/\omega_h$*. Hence*

$$t^{1-g} Z(t, C/k) = \left(\frac{1}{qt}\right)^{1-g} Z\left(\frac{1}{qt}, C/k\right).$$

Sketch of proof : One uses the Riemann–Roch theorem on the curve C of genus g, namely the dimension $l(\mathfrak{a})$ of the k-vector space $L(\mathfrak{a})$ of functions f with poles on a divisor $\mathfrak{a}$ is given by

$$l(\mathfrak{a}) = \deg(\mathfrak{a}) + 1 - g + l(\mathfrak{k} - \mathfrak{a})$$

where $\mathfrak{k}$ is a certain divisor (the canonical divisor) on C, defined over k, of degree $\deg(\mathfrak{k}) = 2g - 2$.

Of course, $l(\mathfrak{a}) = 0$ if $\deg(\mathfrak{a}) \le 0$, unless $\mathfrak{a} = 0$ in which case $L(\mathfrak{a})$ consists of constants and $l(\mathfrak{a}) = 1$.

Setting $\mathfrak{a} = \mathfrak{k}$ one gets $l(\mathfrak{k}) = g$.

If $g = 0$ it is immediate that $Z(t, \mathbb{P}^1/k) = 1/((1-t)(1-qt))$. Suppose $g \ge 1$. In order to count the number of effective divisors of degree n, one notes that divisors of given degree n split into finitely many classes, say $h(C)$, up to equivalence $\mathfrak{a} = \mathfrak{b} + (f)$ where f is a rational function. By the Riemann–Roch theorem, the number of divisors of degree n in a fixed equivalence class is, if $n > 2g-2$, the same as the cardinality of $(L(\mathfrak{a}) - \{0\})/k^*$, hence it is $(q^{n+1-g} - 1)/(q-1)$. Therefore, the associated generating function $R(t)$ is

$$R(t) := \sum_{\deg(\mathfrak{a}) > 2g-2} t^{\deg(\mathfrak{a})} = h(C) \sum_{n=2g-1}^{\infty} \frac{q^{n+1-g} - 1}{q-1}\, t^n,$$

easily seen to be a rational function of t with denominator $(q-1)(1-qt)$.

Hence

$$Z(t, C/k) = \sum_{n=1}^{2g-2} \sum_{\deg(\mathfrak{a})=n}{}' \frac{q^{l(\mathfrak{a})} - 1}{q-1} t^n + R(t)$$

where $\sum'$ is over inequivalent classes and $R(t)$ is the rational function described above.

Another equation for $Z(t, C/k)$ is obtained using in the first piece the Riemann–Roch equation $l(\mathfrak{a}) = \deg(\mathfrak{a}) + 1 - g + l(\mathfrak{k} - \mathfrak{a})$ and the computation of $\deg(\mathfrak{k})$ and $l(\mathfrak{k})$. Comparison of the two equations yields, after a simple calculation, the desired functional equation

$$t^{1-g} Z(t, C/k) = \Big(\frac{1}{qt}\Big)^{1-g} Z\Big(\frac{1}{qt}, C/k\Big).$$

In terms of the 'old' zeta function $\zeta(s, C/k)$ this can be written as

$$q^{-s(g-1)} \zeta(s, C/k) = q^{-(1-s)(g-1)} \zeta(1-s, C/k)$$

and the parallel with the functional equation of the Dedekind zeta function of a number field becomes evident.

The Riemann hypothesis for $\zeta(s, C/k)$ is the statement that all its zeros have real part exactly $\frac{1}{2}$. Equivalently, the numbers ω_h (the so-called characteristic roots) satisfy

$$\textit{Riemann hypothesis:} \qquad |\omega_h| = \sqrt{q}.$$

Note that since the rational function $Z(t, C/k)$ is defined over $\mathbb{Q}$, the same property holds for all algebraic conjugates of ω_h, that also must be characteristic roots. Since the leading coefficient of the infinite series for $Z(t, C/k)$ is 1, the characteristic roots are algebraic integers.

An algebraic integer whose algebraic conjugates all have the same absolute value $(\sqrt{q})^m$, with $m \in \mathbb{N}$, is called a *Weil number*.

Central to the proof is the Explicit Formula which in this case may be written as

$$|C(k_\nu)| = 1 - \sum_{h=1}^{2g} \omega_h^\nu + q^\nu.$$

This is obtained by taking the logarithmic derivative of $Z(t, C/k)$.

On the other hand, rational points on C/k_ν are simply the points of C which are invariant by the Galois group of $\overline{k}/k_\nu$, which is generated by the ν-th power of the Frobenius automorphism $F(x) = x^q$, i.e. $F^\nu(x) = x^{q^\nu}$. Hence if we have a good cohomology theory available for the curve C the Lefschetz fixed point formula reads

$$|C(k_\nu)| = \sum_{i=0}^{2} (-1)^i \operatorname{Trace}\big(F^\nu | H^i(C, *)\big)$$

where the $*$ stands for an appropriate coefficient system.

The cohomology theory in question was constructed by Grothendieck and his collaborators as the l-adic étale cohomology with $\mathbb{Q}_l$ coefficients, where l is a prime number unequal to p. The good topology is the *étale topology* where open sets are étale (unramified) covers of Zariski open sets. The computation of traces in the l-adic cohomology yields

$$\operatorname{Trace}\big(F^\nu | H^0(C, \mathbb{Q}_l)\big) = 1, \quad \operatorname{Trace}\big(F^\nu | H^2(C, \mathbb{Q}_l)\big) = q^\nu$$

hence we must have

$$\operatorname{Trace}\big(F^\nu \mid H^1(C, \mathbb{Q}_l)\big) = \sum_{h=1}^{2g} \omega_h^\nu.$$

In other words,

$$P(t, C/k) = \det(I - (F | H^1(C, \mathbb{Q}_l))t)$$

is the characteristic polynomial of F relative to its action on the first cohomology group H^1 of the curve C.

In fact, the l-adic absolute value of any characteristic root is 1 (i.e. they are units), while the p-adic absolute value need not be 1.

Serre has shown that a good cohomology theory with real coefficients for varieties over a field of characteristic p does not exist. This explains in part why it is hard to fit the Riemann hypothesis into a nice formal setting.

There is however a purely geometric interpretation of the Explicit Formula, based on the theory of algebraic correspondences. Using the theory of correspondences, Weil around 1940 obtained the long-awaited proof of the Riemann hypothesis for the Zeta function of a curve over a finite field. In the next section, we will describe Weil's proof.

18. The Riemann hypothesis: Correspondences

An effective correspondence between two curves C and C' is an algebraic multivalued map from C to C'. Thus it may be viewed as a morphism $C \to S^d C'$ where S^d is the *d-fold symmetric product* of C'. Another way of looking at a correspondence is to consider its graph on the product $C \times C'$, which is a divisor on the surface $C \times C'$. This allows one to consider 'virtual' correspondences, identifying them with a divisor (not necessarily effective) on $C \times C'$.

The most important case occurs when $C = C'$. Correspondences from C to C form an associative ring, with addition given by the sum of divisors on $C \times C$ and product given by *composition*: If X and Y are two correspondences, we write $X \circ Y$ for their composition as multivalued mappings $C \to C$.

The *diagonal* Δ of $C \times C$ is the identity element of the ring of correspondences.

The ring of correspondences has an obvious involution $X \to \widetilde{X}$ obtained by exchanging the two factors of $C \times C$.

Let X be an effective correspondence $P \mapsto \{X_1(P), \dots, X_m(P)\}$, transversal with the diagonal Δ. Then:

- The number of fixed points $P = X_i(P)$ equals the intersection number (X, Δ) on the surface $C \times C$.
- The *degree* $d(X) := m$ equals the intersection number $(X, P \times C)$.
- The *dual degree* $\widetilde{d}(X) := d(\widetilde{X})$ equals the intersection number $(X, C \times P)$.

These concepts are extended to arbitrary correspondences by linearity.

If F is the graph of the Frobenius correspondence $C \times C$, then

$$d(F) = 1, \quad \widetilde{d}(F) = q, \quad (F, \Delta) = |C(k)|$$

(The equation $x = a^q$ has one solution, the equation $x^q = a$ has q solutions, the equation $x^q = x$ characterizes rational points.)

Thus $T[F] = d(F) + \widetilde{d}(F) - (F, \Delta)$. In general, the *trace* of a correspondence is by definition

$$T[X] := d(X) + \widetilde{d}(X) - (X, \Delta).$$

If $C = \mathbb{P}^1$ (i.e. has genus 0) then $T[X] = 0$ for any X. The 'old masters' called $T[X]$ the *defect* of the correspondence.

Theorem. *Let X be a correspondence on a projective non-singular curve C. Then*

$$T[X \circ \widetilde{X}] = d(X \circ \widetilde{X}) + \widetilde{d}(X \circ \widetilde{X}) - (X \circ \widetilde{X}, \Delta) \geq 0.$$

Steps of the proof (Weil, simplified by Swinnerton-Dyer): Define

$$\langle X, Y \rangle = d(X \circ \widetilde{Y}) + \widetilde{d}(X \circ \widetilde{Y}) - (X \circ \widetilde{Y}, \Delta).$$

Then

- The left- and right-kernel of $\langle X, Y \rangle$ consist of the space of correspondences spanned by $P \times C$, $C \times P$, and the divisors on $C \times C$ numerically equivalent to 0.

- If $d(X) \geq g$, there is a rational function f on $C \times C$ such that $X + (f)$ is effective. For the proof, use the Riemann–Roch theorem on C: The general divisor on C given by $X(P)$ can be changed, by a function f_P on C, to an effective divisor $X(P) + (f_P)$. Then define $f(P, Q) = f_P(Q)$.

Since adding a multiple of $P \times C$ to X does not change the trace, we may assume $d(X) = g$.

If $g = 1$, then $\langle X, X \rangle = 2\widetilde{d}(X) \geq 0$ (X is effective!) and there is nothing more to prove. This yields the Riemann hypothesis for elliptic curves over a finite field and is equivalent to one of two proofs by Hasse (1933) in this case. However, until Weil's work it remained unclear how to extend Hasse's argument to curves of arbitrary genus.

If $g \geq 2$, write $X(P) = \{X_1(P), \ldots, X_g(P)\}$; then $X \circ \widetilde{X}$ is the locus, as P varies, of all the points $X_i(P) \times X_j(P)$. The terms with $i = j$ contribute $\widetilde{d}(X)\Delta$, because each $X_j(P)$ arises from precisely $\widetilde{d}(X)$ points. Removing this component from $X \circ X'$ we remain with a divisor Y, defined over k, and

$$\langle X, X \rangle = 2g\widetilde{d}(X) + (2g-2)\widetilde{d}(X) - (Y, \Delta)$$

because $(\Delta, \Delta) = 2 - 2g$.

It remains to compute an upper bound for (Y, Δ), i.e. for the number of coincidences (P, i, j) for which $(X_i(P), X_j(P)) \in \Delta$ with $i \neq j$, i.e. $X_i(P) = X_j(P)$, $i \neq j$.

The idea of how to compute such a bound is very clever. Let $\mathfrak{k}$ be an effective canonical divisor of C, let f_i, $i = 1, \ldots, g$, be a basis of rational functions on C such that $\mathfrak{k} + (f_i) > 0$, and consider the function

$$\Phi(P) := \det \Big(f_i(X_j(P)) \Big)^2.$$

Then $\Phi(P)$ is a rational function defined over k, because a permutation of the points $X_j(P)$ and an automorphism of $\overline{k}/k$ only permutes the columns of the determinant thus, by squaring, there is no change in sign. If $X_j(P) = X_{j'}(P)$ then two columns are identical and we get a zero of $\Phi(P)$, thus $(Y, \Delta) \leq \deg(\Phi(P))$, i.e. the number of poles. The poles of Φ are double poles at those points P for which $X_j(P)$ is a component of $\mathfrak{k}$, giving $(2g-2)\widetilde{d}(X)$ double poles. Hence $(Y, \Delta) \leq (4g-4)\widetilde{d}(X)$ and from

$$\langle X, X \rangle = 2g\widetilde{d}(X) + (2g-2)\widetilde{d}(X) - (Y, \Delta)$$

we see that $\langle X, X \rangle \geq 2\widetilde{d}(X) \geq 0$, proving the theorem.

This is the famous *defect inequality* for a correspondence. It deserves, together with the algebraic index theorem for surfaces, a short historical note.

The foundation of a general theory of correspondences between curves was done by Severi [49] in 1903 and he conjectured the defect inequality in that paper. In 1906, he was able to obtain an important theorem on the quadratic intersection form for

curves on a projective non-singular algebraic surface, showing that if two curves C and D on a projective non-singular surface satisfy the relation

$$(C \cdot C) = (C \cdot D) = (D \cdot D) > 0$$

then a positive integral multiple of $C - D$ had to be algebraically equivalent to 0.[9] This was his well-known criterion for algebraic equivalence, up to torsion, of curves on a surface. In 1906 again, Castelnuovo observed that if the surface was a product $C \times C$ then Severi's theorem implied the conjectured defect inequality, giving only a quick sketch of the argument. In 1926, Severi gave a complete algebraic proof of the defect inequality. A little later, Castelnuovo [16] obtained an elegant new transcendental proof (hence limited to characteristic 0) of Severi's inequality. He worked with the space of holomorphic differential forms on the curve and the endomorphism of the associated vector space induced by the correspondence, in what today would be considered to be a cohomological proof. Castelnuovo's proof became the standard and the original proof by Severi was forgotten.

A similar path occurred for the algebraic index theorem for surfaces. In 1930, Hodge proved his famous index theorem for harmonic cohomology classes of a projective variety of even dimension; in particular, the quadratic intersection form for divisors on a surface has signature $(+, -, \dots, -)$. Hodge's fundamental work attracted a lot of attention and in 1937 Segre gave a simple algebraic proof of the algebraic index theorem for surfaces based on, and credited to, Severi's original idea of 1906.[10] In 1939 Bronowski independently rediscovered this proof, setting it in a more modern form, with credits to Hodge and du Val. In 1958 Mattuck and Tate, unaware of these earlier works, rediscovered Bronowski's argument and also referred to a forthcoming paper by Grothendieck for new light on the subject. Grothendieck's paper [33], at last, gave credit to both Segre and Bronowski for providing simple proofs in arbitrary characteristic and also contained some new applications. His proof, like all known algebraic proofs of this theorem, was once again based on the Riemann–Roch theorem for a surface, with the same basic argument.

It was Weil that realized for the first time that, lacking a good cohomology theory, the theory of correspondences could be used instead for attacking the Riemann hypothesis for curves. Around 1940 he was able to extend Severi's algebraic proof of the defect inequality to the all-important case of curves over a finite field; in order to do this in a proper fashion, he first had to lay down the foundations of algebraic geometry over fields of positive characteristic, which was never considered by the Italian school of algebraic geometry. In his short announcement paper [57], Weil briefly acknowledged in a reluctant fifth footnote the existence of an algebraic

[9] In modern language, this states that the self-intersection quadratic form of integral divisors on a surface restricted to the orthogonal complement of a divisor with positive self-intersection represents 0 only on torsion divisors. Severi goal was to characterize algebraic equivalence on a surface. What he proved there was that numerical equivalence on a surface coincides with algebraic equivalence, up to torsion.

[10] As yet, no algebraic proof of an algebraic index theorem for smooth projective varieties of even dimension larger than 2 has been found; the problem remains wide open for varieties in positive characteristic.

proof in characteristic 0 due to Severi, stressing the fact that one could not quite appeal to that result because of the need of working in positive characteristic.[11] The war delayed the publication of the proof and Weil resorted to sending some details of his arguments in letters to Artin and to his sister Simone, fearing that because of the raging war they might otherwise be lost. Eventually, the full proof was published in 1948 in a landmark memoir [59].

It remains to apply the defect inequality to our situation. We apply it to the correspondence $m\Delta + nF_\nu$. Then

$$\begin{aligned}\langle m\Delta + nF_\nu, m\Delta + nF_\nu\rangle &= \langle \Delta, \Delta\rangle m^2 + 2\langle \Delta, F_\nu\rangle mn + \langle F_\nu, F_\nu\rangle n^2 \\ &= 2gm^2 + 2(1 + q^\nu - |C(k_\nu)|)mn + 2gq^\nu n^2 \geq 0\end{aligned}$$

and in particular its discriminant is non-negative. Hence

$$\left(1 + q^\nu - |C(k_\nu)|\right)^2 - (2g)^2 q^\nu \leq 0.$$

Thus, by the Explicit Formula we find

$$\Big|\sum_{h=1}^{2g} \omega_h^\nu\Big| \leq 2g(\sqrt{q})^\nu, \quad \nu = 1, 2, \ldots$$

and it follows easily that $|\omega_h| \leq \sqrt{q}$. For a quick proof, one may note that the power series

$$\sum_{h=1}^{2g} \frac{1}{1 - \omega_h z} = \sum_{\nu=0}^{\infty} \left(\sum_{h=1}^{2g} \omega_h^\nu\right) z^\nu$$

will have radius of convergence at least $1/\sqrt{q}$, which implies $|\omega_h| \leq \sqrt{q}$.

Finally, the functional equation shows that also $|q/\omega_h| \leq \sqrt{q}$, proving the equality $|\omega_h| = \sqrt{q}$ and the Riemann hypothesis for the zeta function $\zeta(s, C/\mathbb{F}_q)$.

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